Supplementary Materials

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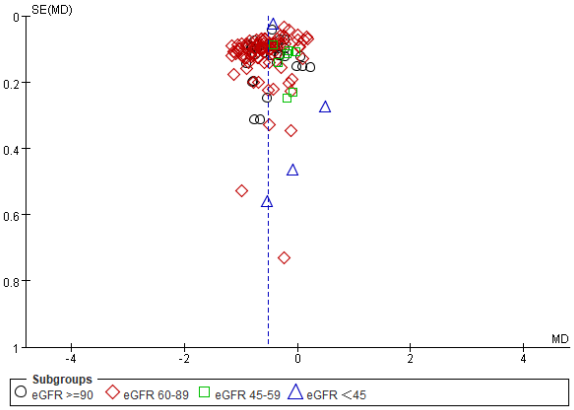
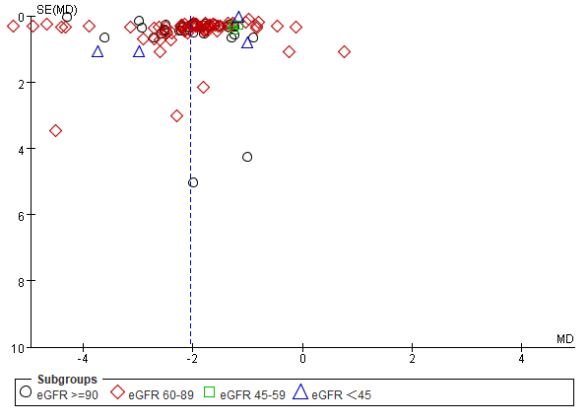
Table S8. Efficacy of SGLT2i treatment in patients with different levels of renal function (follow-up period ⩾12 weeks).

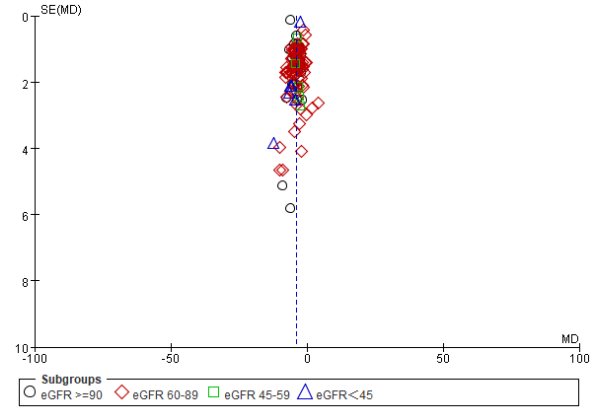
Figure S1. Funnel plots of analyses for efficacy and safety of SGLT2is.



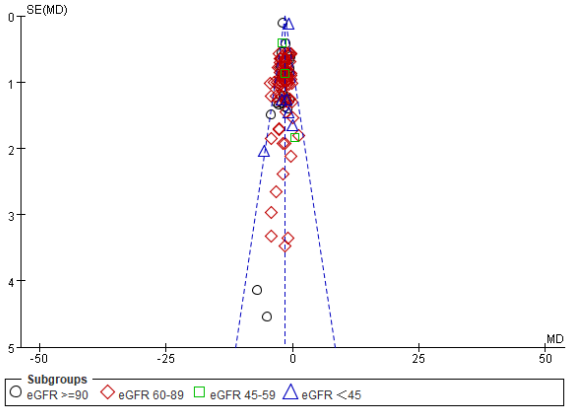
(a) HbA1c changes



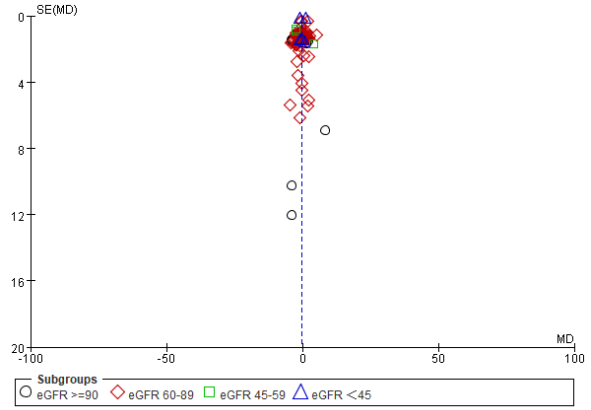
(b) Weight changes



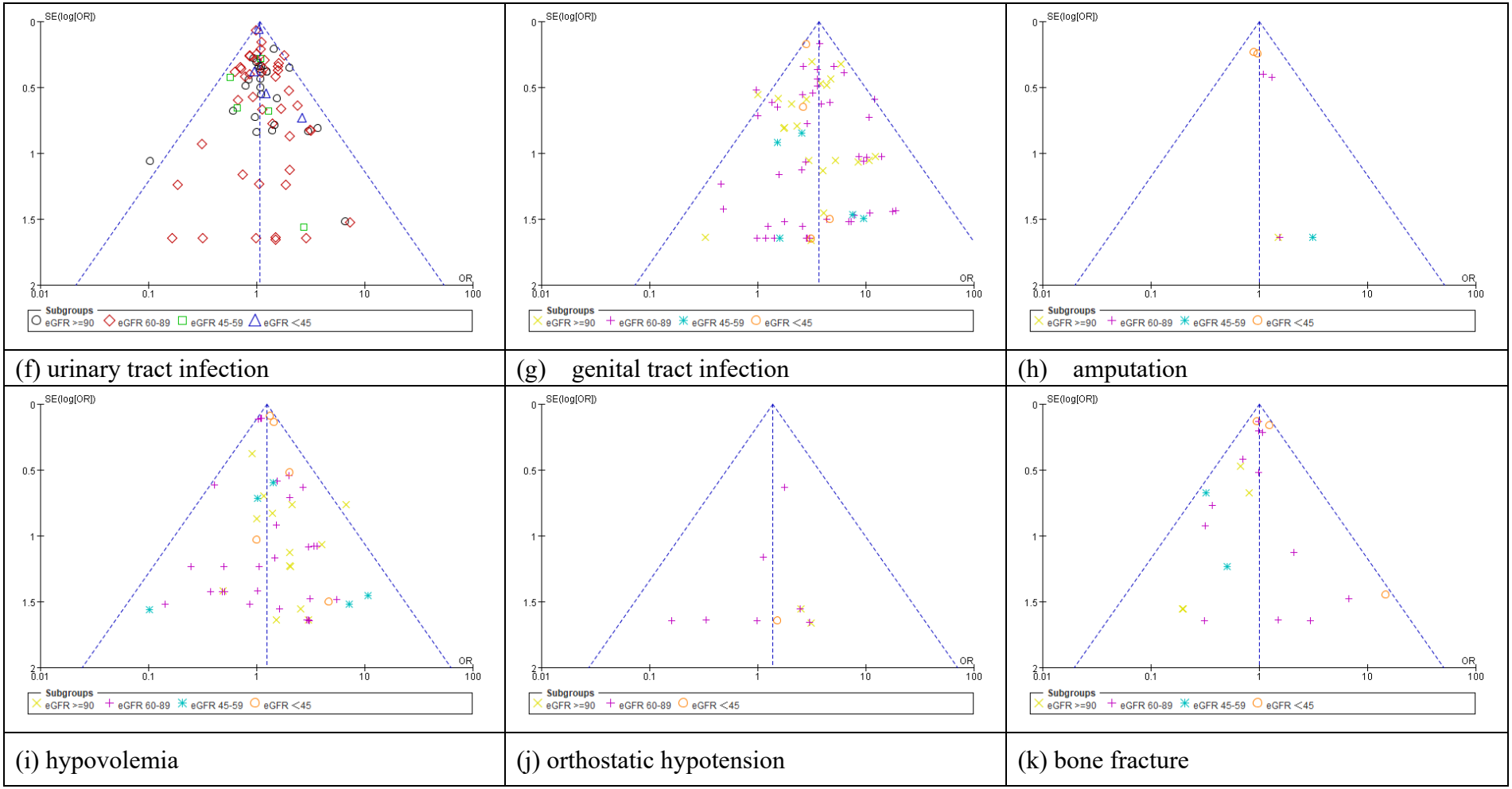
(c) SBP changes



(d) DBP changes



(e) eGFR changes



(f) urinary tract infection

(g) genital tract infection

(h) amputation

(i) hypovolemia

(j) orthostatic hypotension

(k) bone fracture

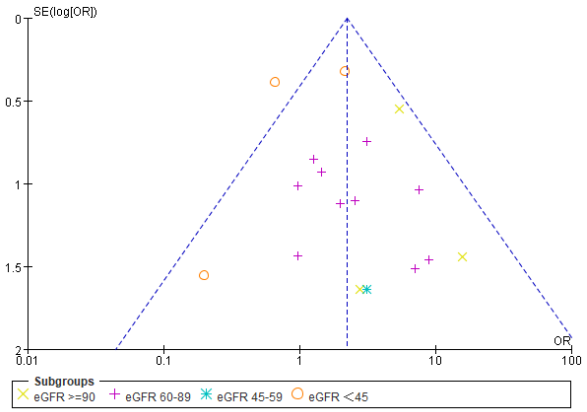
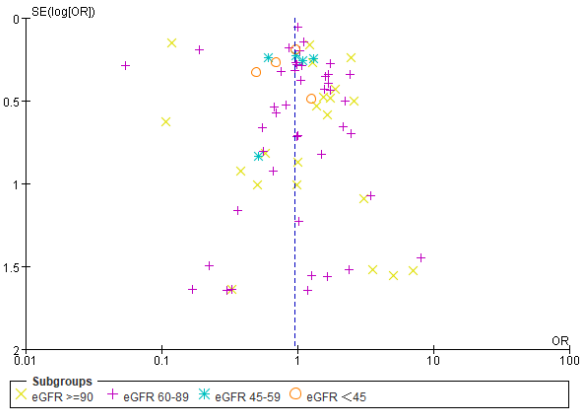
		
(l) diabetic ketoacidosis(DKA)	(m) hypoglycemia	

Figure S2. HbA1c changes before and after SGLT2i treatment in patients with different levels of renal function.

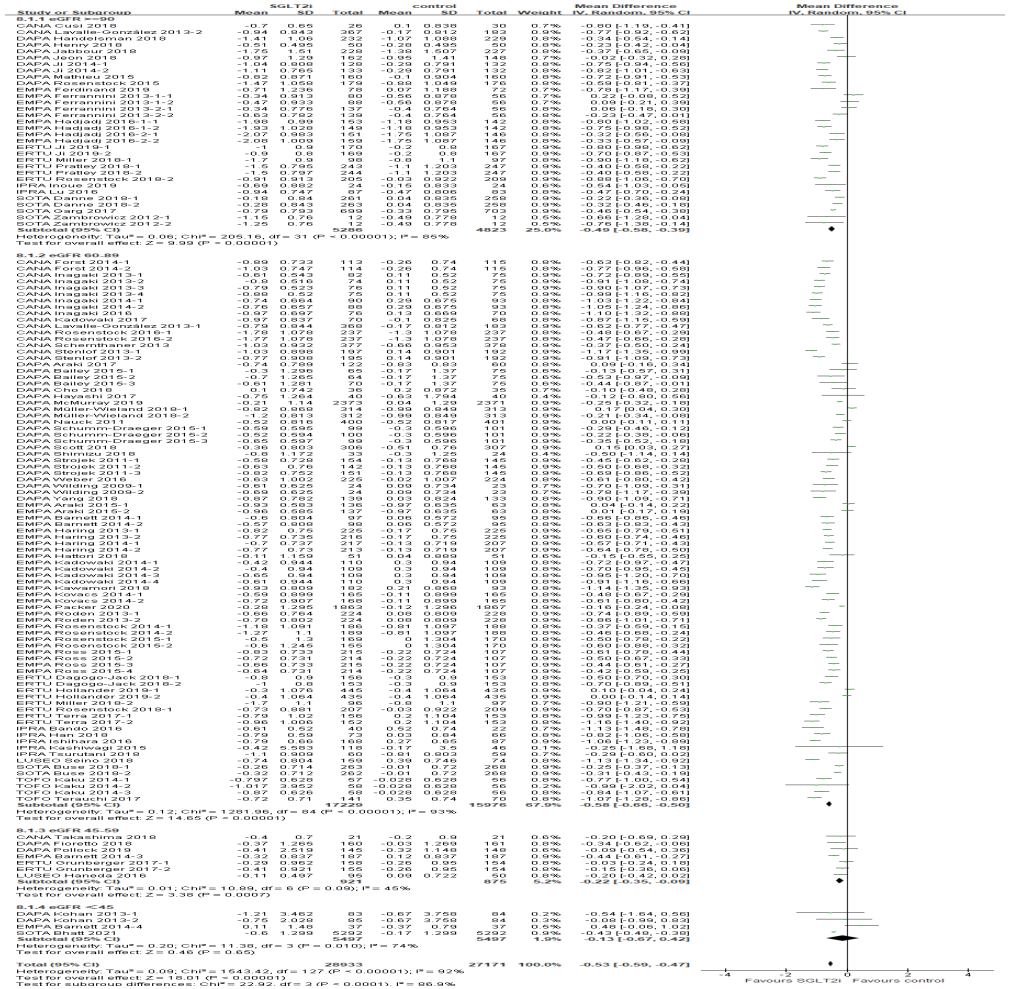


Figure S3. Weight changes before and after SGLT2i treatment in patients with different levels of renal function.

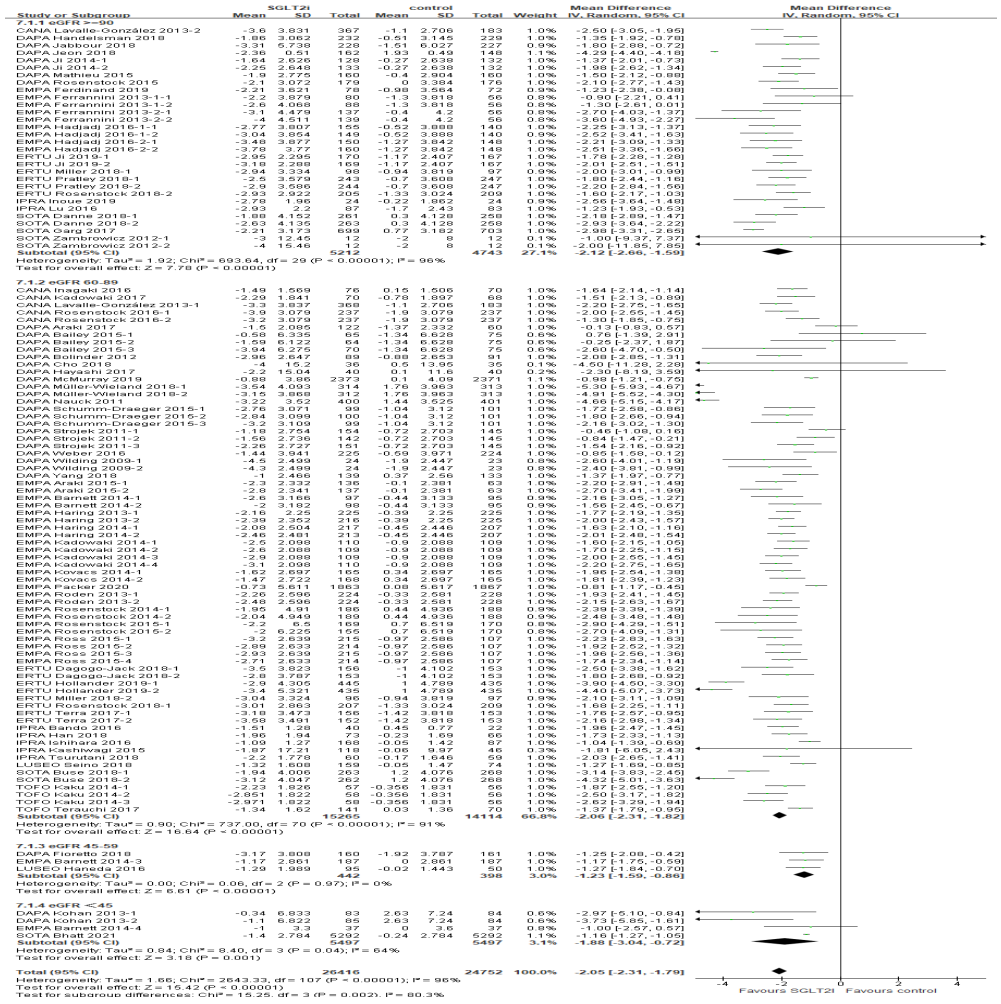


Figure S4. SBP changes before and after SGLT2i treatment in patients with different levels of renal function.

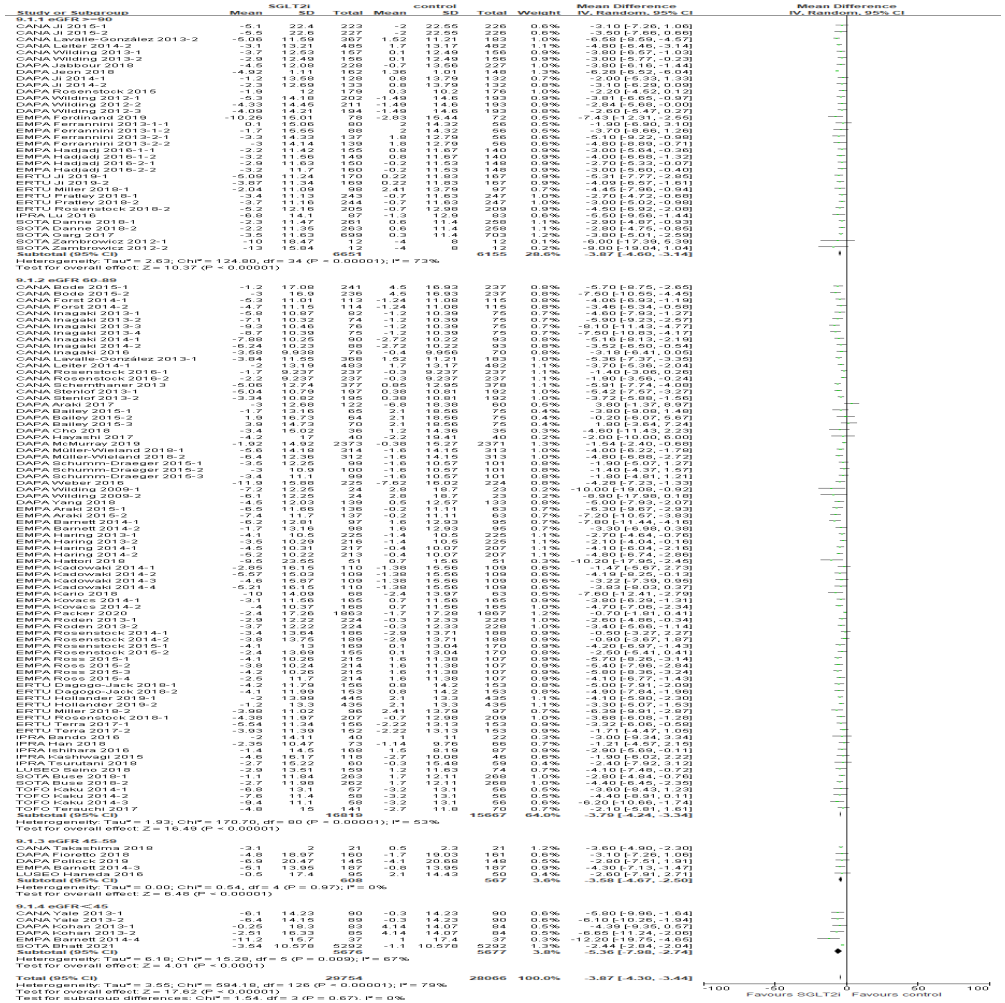


Figure S5. DBP changes before and after SGLT2i treatment in patients with different levels of renal function.

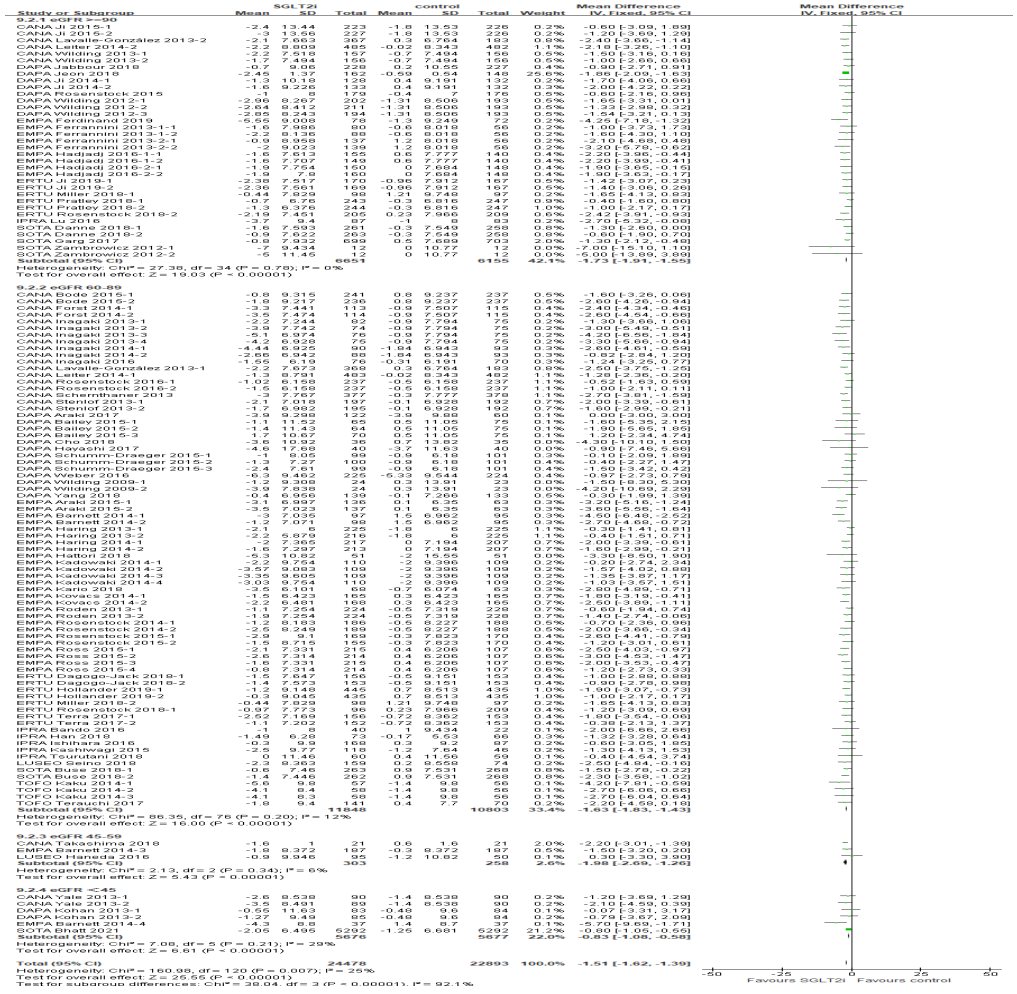


Figure S6. eGFR changes before and after SGLT2i treatment in patients with different levels of renal function.

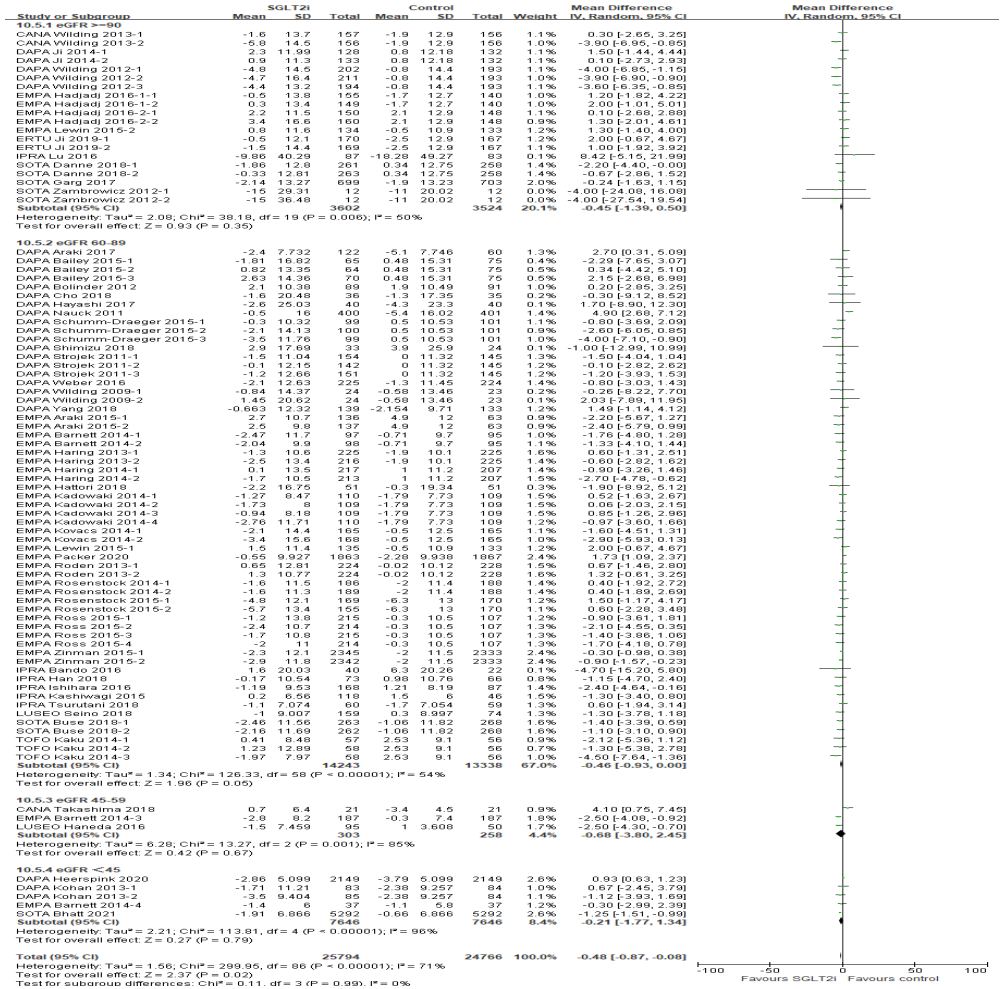


Figure S7. The associations between the use of SGLT2i treatment and urinary tract infection.

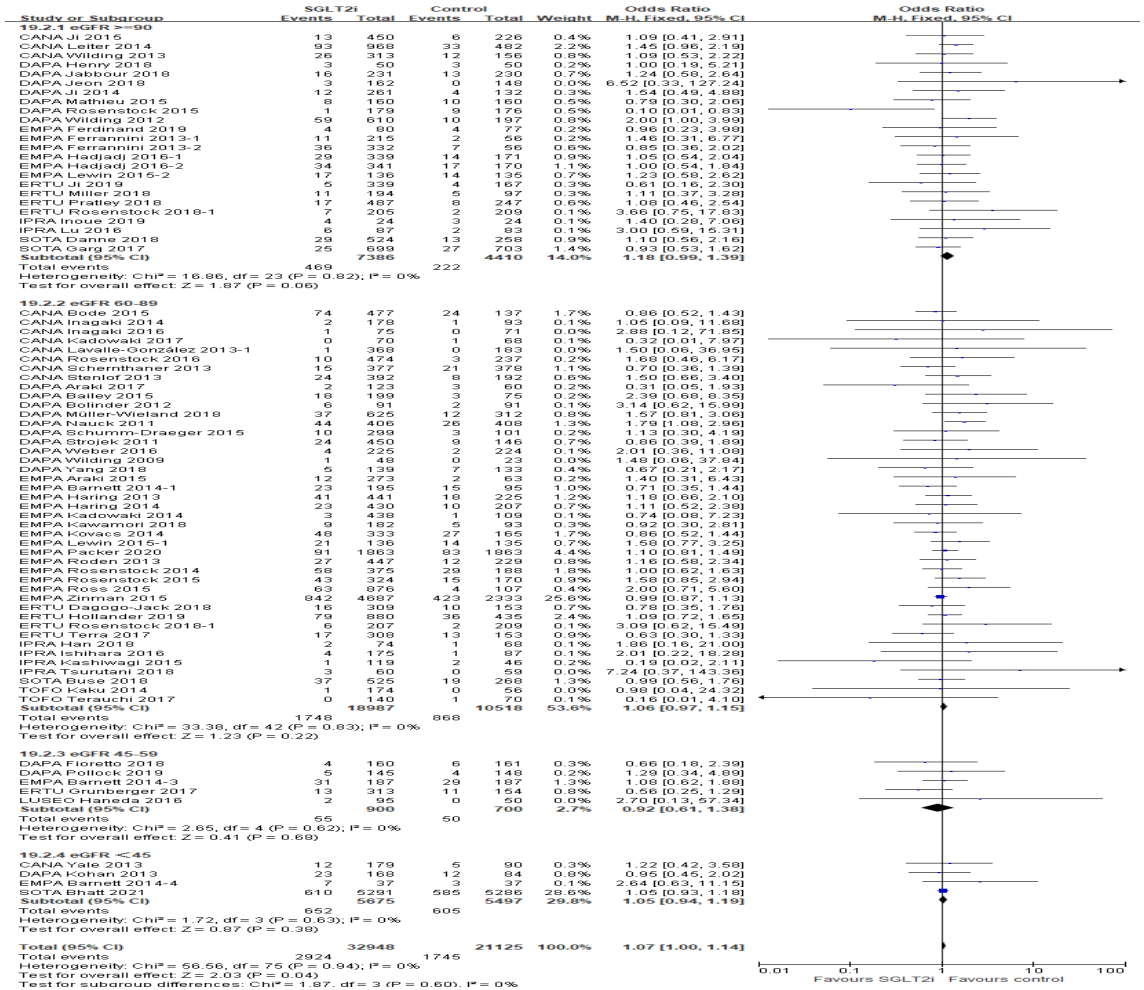


Figure S8. The associations between the use of SGLT2i treatment and genital tract infection.

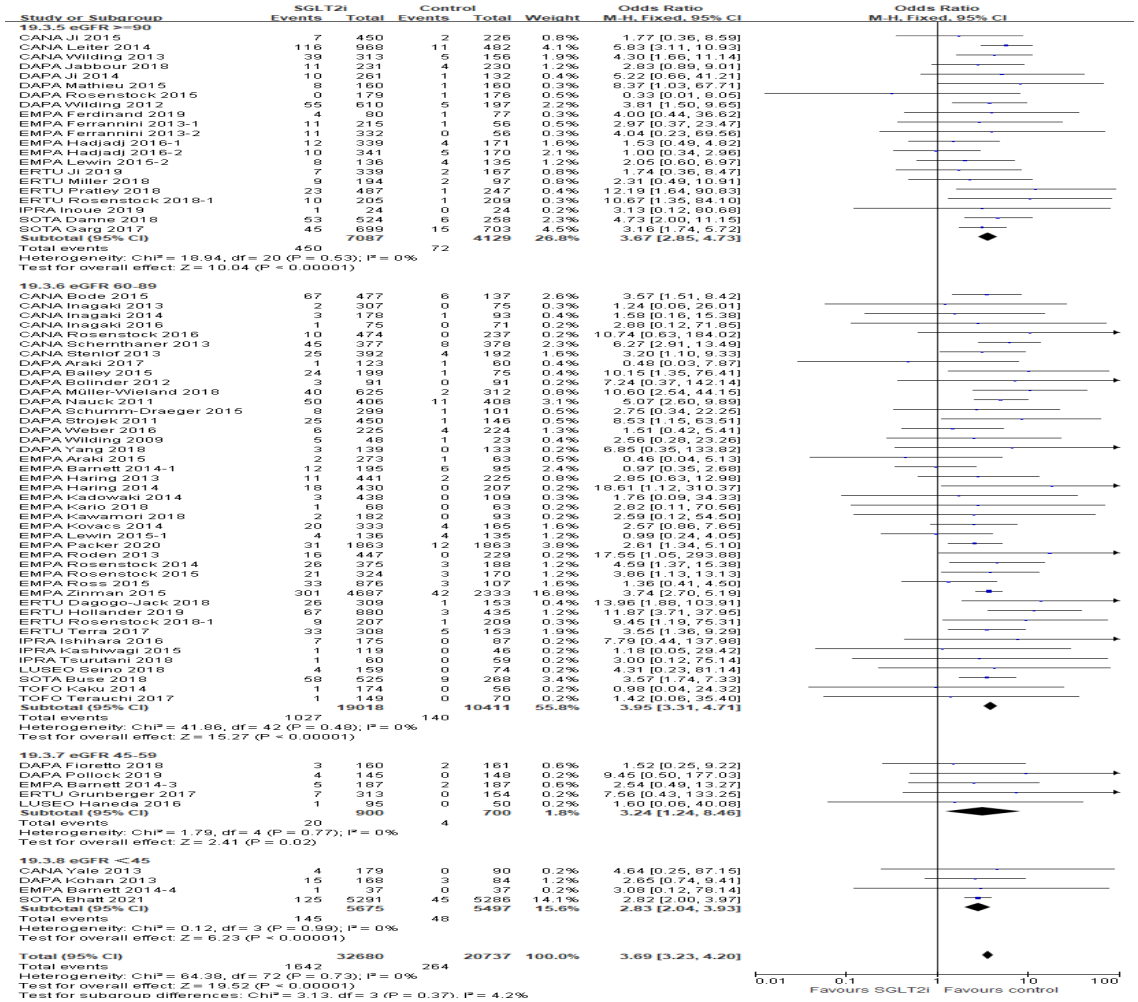


Figure S9. The associations between the use of SGLT2i treatment and amputation.

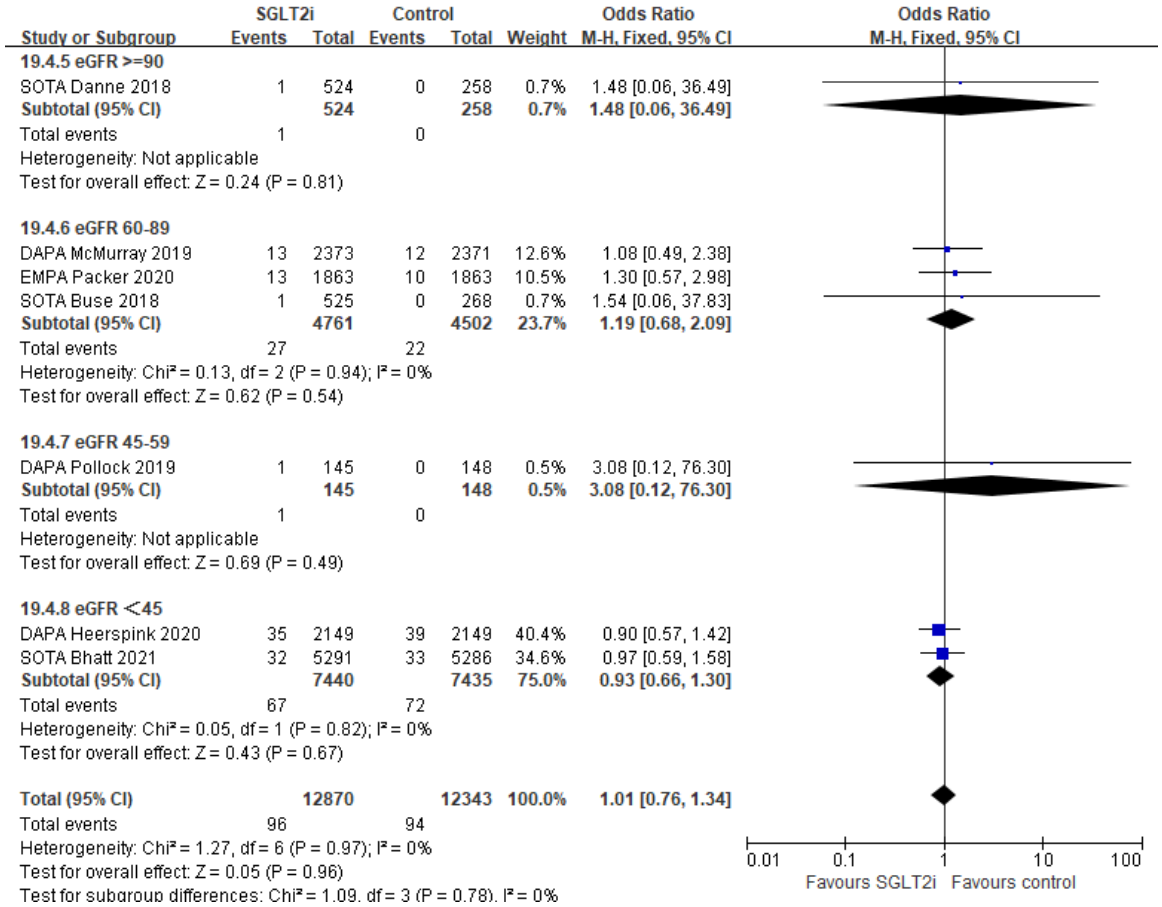


Figure S10. The associations between the use of SGLT2i treatment and hypovolemia.

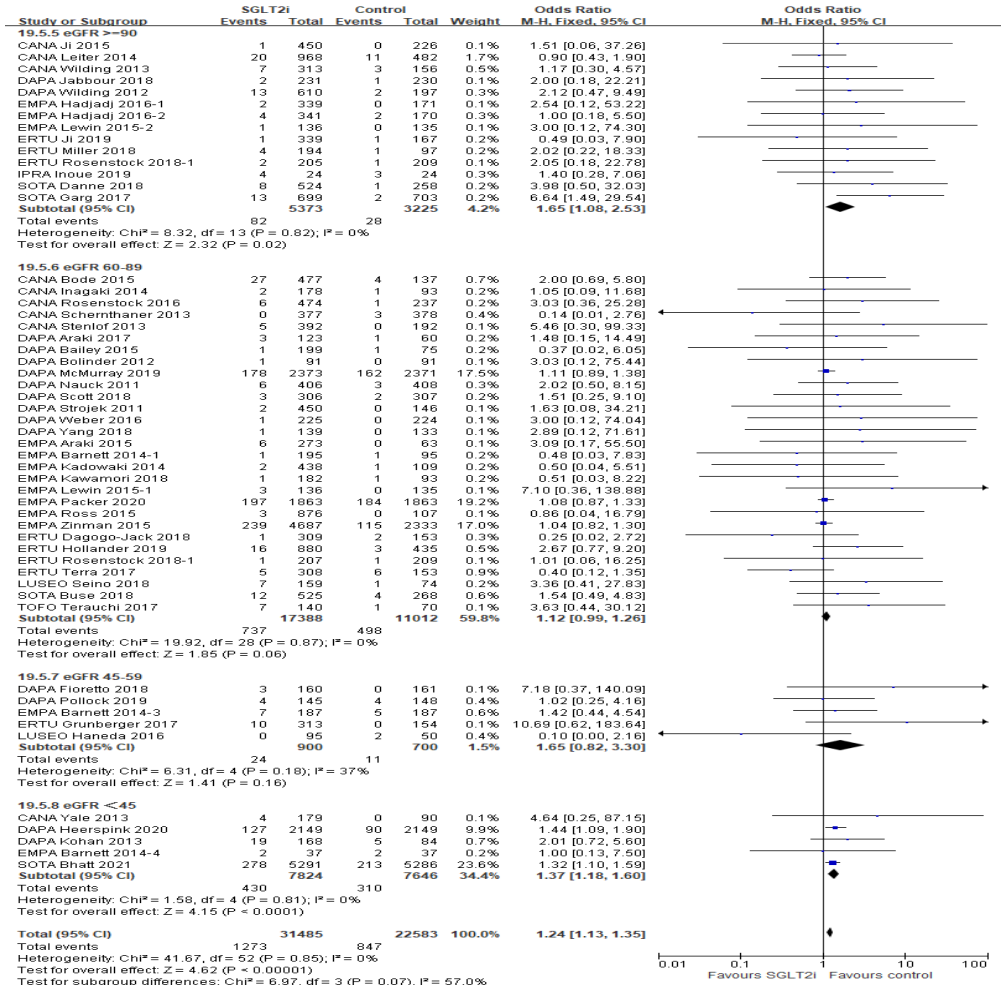


Figure S11. The associations between the use of SGLT2i treatment and orthostatic hypotension.

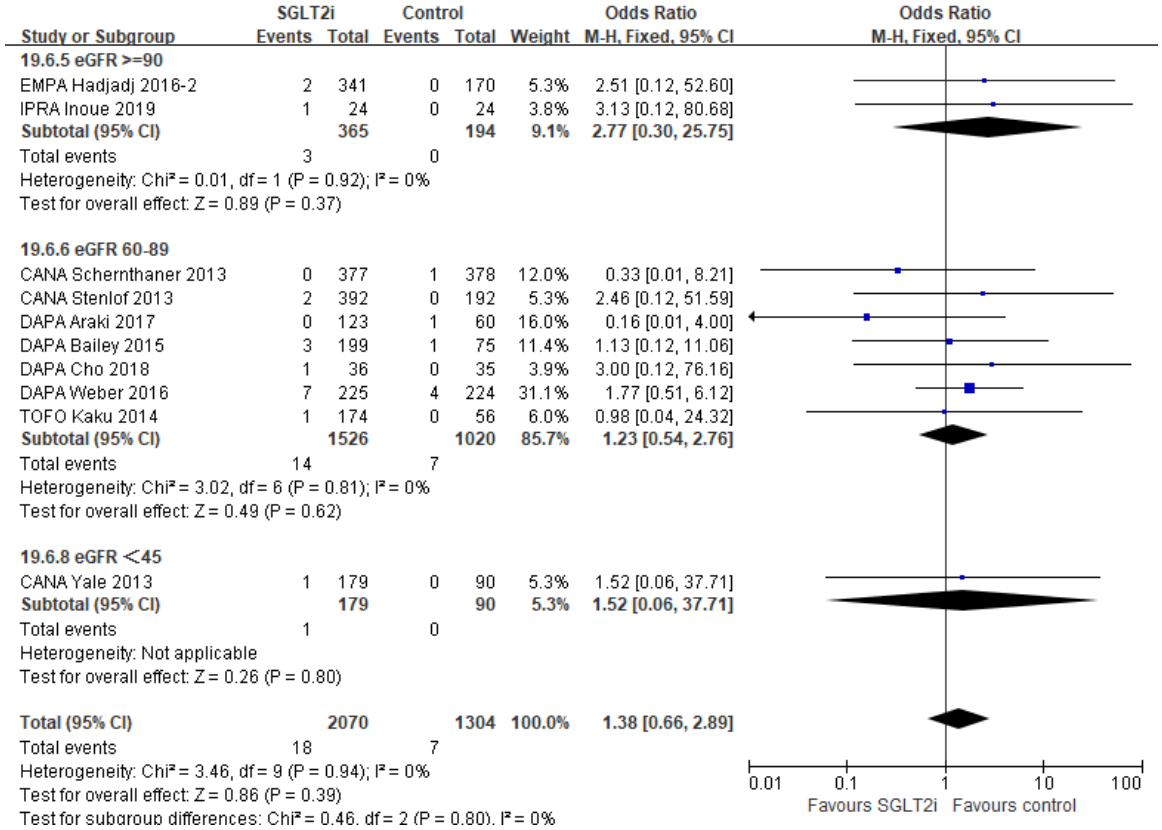


Figure S12. The associations between the use of SGLT2i treatment and bone fracture.

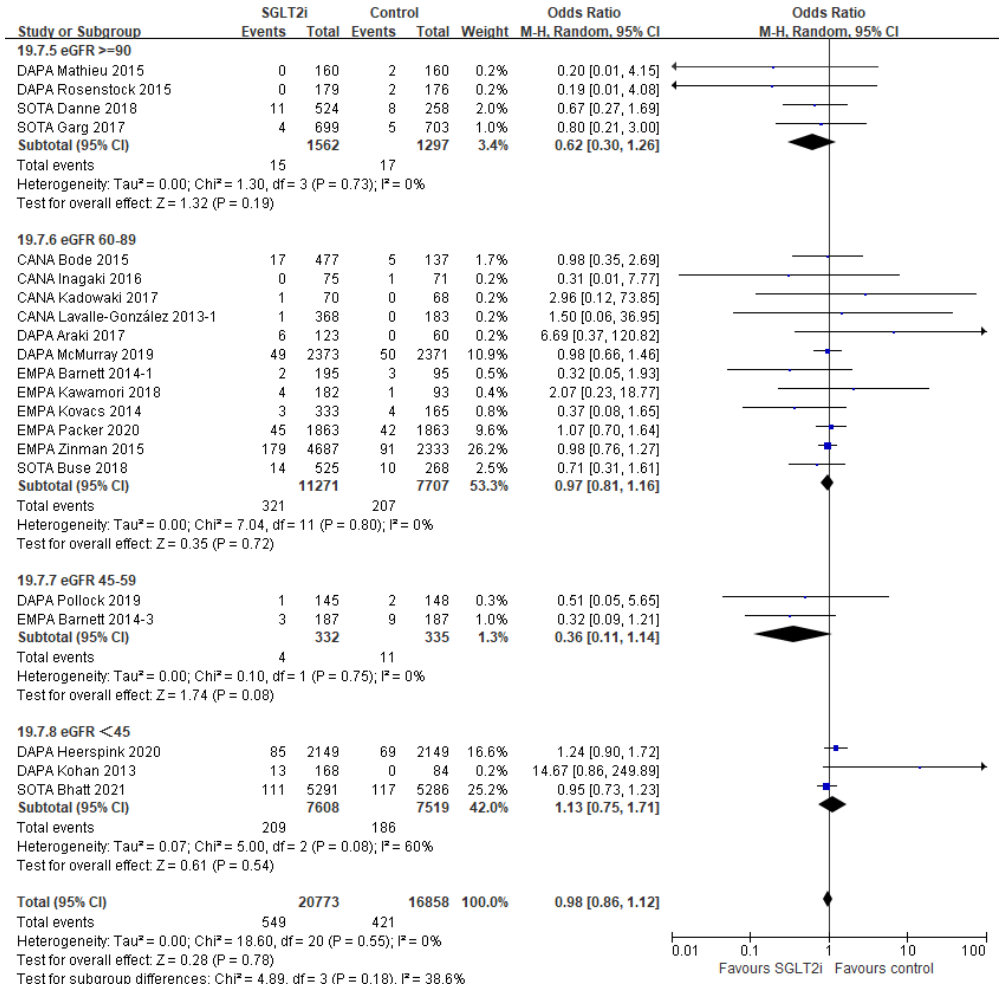


Figure S13. The associations between the use of SGLT2i treatment and diabetic ketoacidosis (DKA).

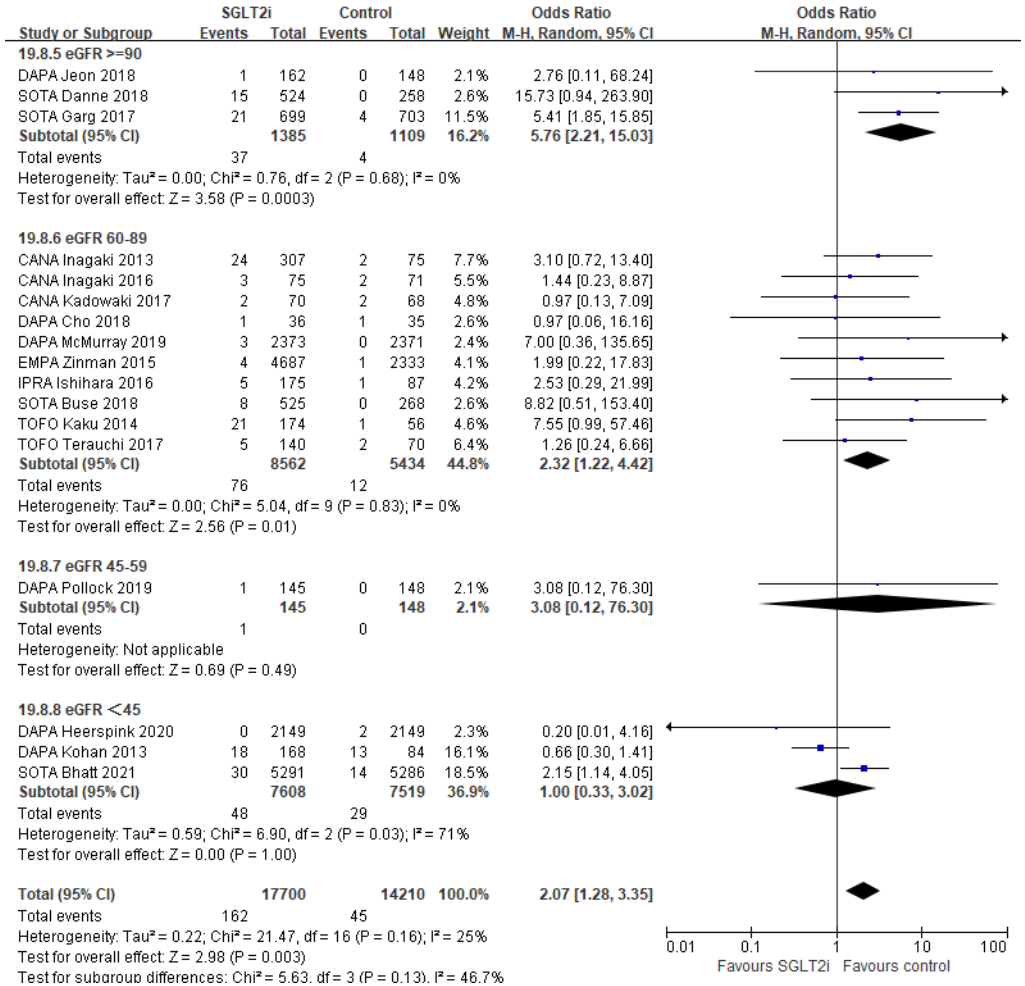


Figure S14. The associations between the use of SGLT2i treatment and hypoglycemia.

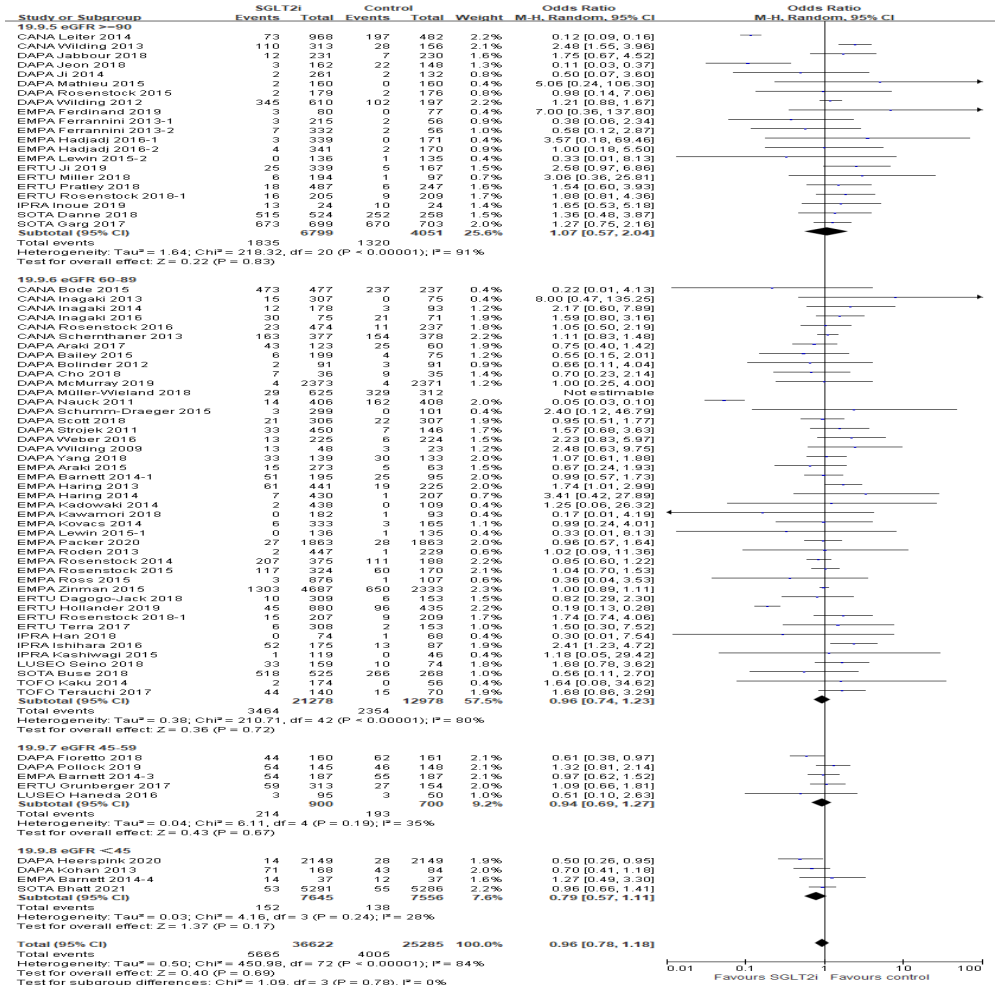


Table S1. Baseline characteristics of included RCTs.

First author, year	Condition	Study duration(w)	Treatment group	Dosage	No. of patients	Age(year)	Male percentage (%)	Baseline eGFR(ml/min/1.73m^2)
baseline eGFR >= 90 ml/min/1.73m^2								
Cusi 2018[1]	T2D	24	cana	300mg	26	58.00	62.00	92.00±16.00
			pbo		30	58.00	70.00	89.00±17.00
Ji 2015[2]	T2D	18	cana	100mg	223	56.50	55.30	92.50±18.30
			cana	300mg	227	56.40	49.80	93.90±18.00
			pbo		226	55.80	55.30	95.40±18.20
Lavalle-González 2013[3]	T2D	26	cana	300mg	367	55.30	45.00	90.20± /
			pbo		183	55.30	51.40	87.7± /
Wilding 2013[4]	T2D	52	cana	100mg qd	157	57.40	48.40	91.00± /
			cana	300mg qd	156	56.10	55.80	91.90± /
			pbo		156	56.80	48.70	87.40± /
Henry 2018[5]	T2D	4	dapa	10mg	50	56.90	52.00	91.40±22.00
			pbo		50	56.80	50.00	89.70±23.60
Jabbour 2018[6]	T2D	52	dapa	10mg	228	53.80	44.70	97.70±23.70
			pbo		227	54.20	51.10	99.40±26.80
Ji 2014[7]	T2D	24	dapa	5mg	128	53.00	65.60	91.60±17.10
			dapa	10mg	133	51.20	64.70	91.70±20.20
			pbo		132	49.90	65.90	94.10±17.70
Mathieu 2015[8]	T2D	24	dapa	10mg	160	55.20	43.70	93.50±20.80
			pbo		160	55.00	47.50	91.60±23.20
Rosenstock 2015[9]	T2D	24	dapa	10mg	179	53.00	47.00	96.60±19.60

			pbo		176	55.00	53.00	92.50±19.50
Wilding 2012[10]	T2D	48	dapa	2.5mg	202	59.80	49.50	111.30±36.00
			dapa	5mg	211	59.30	47.40	109.70±36.70
			dapa	10mg	194	59.30	44.80	114.30±34.90
			pbo		193	58.80	49.20	115.90±39.70
Ferdinand 2019[11]	T2D	24	empa	10mg-25mg	78	56.50	55.10	91.15±18.95
			pbo		72	57.20	50.00	91.49±20.79
Hadjadj 2016[12]	T2D	24	empa	10mg	161	52.20	60.20	93.60±22.30
			empa	25mg	165	51.00	63.60	95.00±20.90
			pbo		168	53.40	51.20	90.80±19.30
			empa	10mg	167	52.30	59.30	93.30±22.00
			empa	25mg	169	53.60	52.10	92.20±19.20
			pbo		164	51.60	56.10	93.40±20.10
Lewin 2015[13]	T2D	52	empa	25mg	134	54.20	52.20	90.10±19.60
			pbo		133	53.80	56.40	89.50±20.30
Ji 2019[14]	T2D	26	ertu	5mg	170	56.10	55.90	97.90±19.20
			ertu	15mg	169	56.30	58.00	100.20±19.80
			pbo		167	56.90	52.70	99.90±20.20
Pratley 2018[15]	T2D	26	ertu	5mg	243	55.20	50.60	91.90±20.40
			ertu	15mg	244	55.10	51.60	92.60±19.20
			pbo		247	54.80	62.30	92.60±18.20
Rosenstock 2018[16]	T2D	26	ertu	15mg	205	56.90	45.40	91.00±20.60
			pbo		209	56.50	46.90	91.60±19.80
Inoue 2019[17]	T2D	24	ipra	50mg	24	60.50	54.20	91.00±26.40

			pbo		24	60.80	58.30	82.90±24.80
Lu 2016[18]	T2D	24	ipra	50mg	87	53.90	50.60	144.12±59.42
			pbo		83	53.40	39.80	154.48±69.97
Danne 2018[19]	T1D	52	sota	200mg	261	42.30	53.30	91.56±18.78
			sota	400mg	263	41.70	50.60	91.66±17.41
			pbo		258	39.70	51.90	92.89±18.25
Garg 2017[20]	T1D	24	sota	400mg	699	43.30	51.20	91.51±19.76
			pbo		703	42.40	48.20	92.47±21.89
Zambrowicz 2012[21]	T2D	4	sota	150mg qd	12	53.00	50.00	129.00±28.00
			sota	300mg qd	12	52.00	66.70	139.00±36.00
			pbo		12	55.00	50.00	114.00±18.00
Leiter 2014[22]	T2D	104	cana	300mg	485	55.80	49.70	91.40±19.40
			glimepiride		482	56.30	54.60	89.50±17.50
Handelsman 2018[23]	T2D	26	dapa+saxagliptin	10mg+5mg	232	55.90	43.10	92.20±20.20
			sitagliptin	100mg	229	55.80	48.00	92.90±22.50
Jeon 2018[24]	T2D	24	dapa	10mg qd	162	59.24	54.32	99.69±10.52
			Insulin glargine		148	56.13	47.30	98.60±12.34
Ferrannini 2013[25]	T2D	90	empa	10mg	80	59.00	46.20	94.40±18.90
			empa	25mg	88	59.00	52.30	92.30±19.10
			metformin		56	58.00	50.00	88.00±19.30
	T2D	90	empa	10mg	137	60.00	50.00	93.60±18.10
			empa	25mg	139	60.00	53.00	92.60±21.40

			sitagliptin		56	60.00	51.80	91.40±17.50
Miller 2018[26]	T2D	26	ertu+sitaglipt in	5mg	98	56.40	58.20	90.00±17.20
			pbo		97	54.30	58.80	92.60±21.60
baseline eGFR [60,90) ml/min/1.73m^2								
Bode 2015[27]	T2D	104	cana	100mg	241	64.30	51.50	77.60±17.00
			cana	300mg	236	63.40	54.70	78.70±16.40
			pbo		237	63.20	60.30	76.10±16.30
Forst 2014[28]	T2D	26	cana	100mg	113	56.70	68.10	84.60±17.50
			cana	300mg	114	57.00	55.30	87.40±19.50
			pbo		115	58.30	66.10	87.20±18.80
Inagaki 2013[29]	T2D	12	cana	50mg	82	57.40	61.00	83.50±16.10
			cana	100mg	74	57.70	70.30	86.90±15.50
			cana	200mg	76	57.00	64.50	83.80±15.00
			cana	300mg	75	57.10	73.30	86.90±15.20
			pbo		75	57.70	72.00	83.00±16.50
Inagaki 2014[30]	T2D	24	cana	100mg	90	58.40	65.60	81.40±13.80
			cana	200mg	88	57.40	81.80	87.20±18.60
			pbo		93	58.20	64.50	84.70±13.70
Inagaki 2016[31]	T2D	16	cana	10mg	76	59.70	57.90	83.80±18.40
			pbo		70	56.10	70.00	86.10±21.70
Kadowaki 2017[32]	T2D	24	cana	100mg	70	58.40	77.10	84.70±15.60
			pbo		68	56.00	77.90	83.90±17.10
Lavalle-González 2013[3]	T2D	26	cana	100mg	368	55.50	47.30	89.70± /

			pbo		183	55.30	51.40	87.70± /
Rosenstock 2016[33]	T2D	26	cana	300mg	237	55.40	48.50	87.00±19.00
			cana	100mg	237	54.20	45.60	89.00±19.00
			pbo		237	55.20	48.90	87.00±19.00
Stenlof 2013[34]	T2D	26	cana	300mg	197	55.30	45.20	86.60±19.10
			cana	100mg	195	55.10	41.50	88.50±20.20
			pbo		192	55.70	45.80	86.00±21.50
Araki 2017[35]	T2D	52	dapa	10mg	122	58.30	73.00	77.30±19.10
			pbo		60	57.60	66.70	79.60±16.40
Bailey 2015[36]	T2D	102	dapa	2.5mg	65	53.00	55.40	85.59±20.13
			dapa	5mg	64	52.60	48.40	83.13±18.43
			dapa	10mg	70	50.60	48.60	85.37±18.95
			pbo		75	52.70	41.30	86.16±17.32
Bolinder 2012[37]	T2D	24	dapa	10mg	89	60.60	55.10	86.00±14.20
			pbo		91	60.80	56.00	82.60±16.10
McMurray 2019[38]	HF	35	dapa	10mg	2373	66.20	76.20	66.00±19.60
			pbo		2371	66.50	77.00	65.50±19.30
Schumm-Draeger 2015[39]	T2D	16	dapa	10mg qd	99	58.50	49.50	82.50±13.82
			dapa	2.5mg bid	100	58.30	37.00	87.70±19.62
			dapa	5mg bid	99	55.30	46.50	86.50±18.10
			pbo		101	58.50	46.50	84.40±18.16
Strojek 2011[40]	T2D	24	dapa	2.5mg	154	59.90	50.00	80.90±18.60
			dapa	5mg	142	60.20	50.00	83.50±19.60
			dapa	10mg	151	58.90	43.70	82.20±17.90

			pbo		145	60.30	49.00	80.20±19.10
Weber 2016[41]	T2D	12	dapa	10mg	225	56.00	52.00	84.80±19.70
			pbo		224	57.00	58.00	87.00±19.50
Wilding 2009[42]	T2D	12	dapa	10mg	24	55.70	54.20	87.20±33.00
			dapa	20mg	24	56.10	54.20	89.20±16.00
			pbo		23	58.40	69.60	86.20±21.50
Yang 2018[43]	T2D	24	dapa	10mg	139	56.50	47.50	88.24±19.36
			pbo		133	58.60	48.10	88.32±19.78
Barnett 2014[44]	T2D	52	empa	25mg	97	62.00	62.90	72.30±11.20
			empa	10mg	98	63.20	61.20	70.80±10.30
			pbo		95	62.60	58.90	71.80±10.20
Haring 2013[45]	T2D	24	empa	10mg	225	57.00	50.00	86.50±21.80
			empa	25mg	216	57.40	53.00	88.30±22.60
			pbo		225	56.90	50.00	86.90±20.10
Haring 2014[46]	T2D	24	empa	10mg	217	55.50	58.00	89.50±19.60
			empa	25mg	213	55.60	56.00	87.70±19.30
			pbo		207	56.00	56.00	89.70±21.40
Hattori 2018[47]	T2D	52	empa	10mg	51	57.40	74.50	74.30±16.30
			pbo		51	58.10	80.40	71.90±18.60
Kario 2018[48]	T2D	12	empa	10mg	68	70.90	52.90	68.40±15.10
			pbo		63	69.30	52.40	70.10±16.20
Kadowaki 2014[49]	T2D	12	empa	5mg	110	57.30	76.40	86.49±17.48
			empa	10mg	109	57.90	70.60	85.76±14.63
			empa	25mg	109	57.20	77.10	85.23±15.84
			empa	50mg	110	56.60	77.30	86.52±15.86

			pbo		109	58.70	73.40	84.61±14.91
Kawamori 2018[50]	T2D	24	empa	10mg	182	60.00	78.00	89.30±18.30
			pbo		93	59.80	77.40	86.30±15.20
Kovacs 2014[51]	T2D	24	empa	10mg	165	54.70	50.30	84.30±20.90
			empa	25mg	168	54.20	50.60	87.40±24.40
			pbo		165	54.60	44.20	85.50±20.10
Lewin 2015[13]	T2D	52	empa	10mg	135	55.20	54.10	87.80±17.70
			pbo		133	53.80	56.40	89.50±20.30
Packer 2020[52]	HF	52	empa	10mg qd	1863	67.20	76.50	61.80±21.70
			pbo		1867	66.50	75.60	62.20±21.50
Roden 2013[53]	T2D	24	empa	10mg	224	56.20	63.00	87.70±19.16
			empa	25mg	224	53.80	65.00	87.54±18.29
			pbo		228	54.90	54.00	86.83±17.94
Rosenstock 2014[54]	T2D	52	empa	10mg	186	56.70	52.00	84.10±17.80
			empa	25mg	189	58.00	44.00	84.40±16.60
			pbo		188	55.30	40.00	83.40±15.40
Rosenstock 2015[55]	T2D	78	empa	10mg	169	58.60	55.00	85.10±22.80
			empa	25mg	155	59.90	60.00	83.30±25.10
			pbo		170	58.10	53.00	85.00±23.00
Ross 2015[56]	T2D	16	empa	12.5mg bid	215	57.60	57.20	88.60±20.00
			empa	25mg qd	214	58.20	53.30	88.90±19.50
			empa	5mg bid	215	58.80	55.80	89.90±22.30
			empa	10mg qd	214	58.50	50.50	89.20±20.60
			pbo		107	57.90	51.40	89.50±18.50

Dagogo-Jack 2018[57]	T2D	52	ertu	5mg	156	59.20	51.90	87.00±17.50
			ertu	15mg	153	59.70	53.60	86.90±15.60
			pbo		153	58.30	65.40	89.90±17.50
Rosenstock 2018[16]	T2D	26	ertu	5mg	207	56.60	46.90	88.90±17.50
			pbo		209	56.50	46.90	91.60±19.80
Terra 2017[58]	T2D	26	ertu	5mg	156	56.80	57.10	88.50±18.40
			ertu	15mg	152	56.20	59.20	88.30±18.00
			pbo		153	56.10	53.60	86.20±19.40
Bando 2016[59]	T2D	12	ipra	50mg	40	54.80	65.00	79.50±14.90
			pbo		22	55.40	63.60	76.50±13.40
Han 2018[60]	T2D	24	ipra	50mg	73	57.62	50.70	89.38±13.61
			pbo		66	57.44	48.50	90.66±17.47
Ishihara 2016[61]	T2D	16	ipra	50mg	168	58.70	62.50	83.98±20.27
			pbo		87	59.20	58.60	80.11±21.94
Kashiwagi 2015[62]	T2D	24	ipra	50mg	118	63.90	78.00	60.30±13.07
			pbo		46	65.70	78.30	62.70±13.13
Seino 2018[63]	T2D	16	luseo	2.5mg	159	57.40	70.40	86.50±19.30
			pbo		74	57.10	68.90	87.90±20.20
Buse 2018[64]	T1D	52	sota	200mg	263	46.60	47.90	87.04±20.18
			sota	400mg	262	46.40	45.80	86.58±18.93
			pbo		268	45.20	51.10	87.58±18.47
Kaku 2014[65]	T2D	24	tofo	10mg	57	58.60	66.70	84.68±20.04
			tofo	20mg	58	56.60	67.20	86.78±19.62
			tofo	40mg	58	57.00	67.20	86.00±18.18

			pbo		56	56.80	66.10	83.78±17.68
Terauchi 2017[66]	T2D	16	tofo	20mg	141	59.10	63.80	79.70±19.80
			pbo		70	56.40	68.60	79.50±17.00
Leiter 2014-2[22]	T2D	104	cana	100ng	483	56.40	52.20	89.70±19.30
			glimepiride		482	56.30	54.60	89.50±17.50
Miller 2018-2[26]	T2D	26	ertu+sitaglipt in	15mg	96	56.10	55.20	89.50±18.10
			pbo		97	54.30	58.80	92.60±21.60
Schernthaner 2013[67]	T2D	52	cana	300mg	377	56.60	54.90	88.30± /
			sitagliptin	100mg	378	56.70	56.90	89.50± /
Nauck 2011[68]	T2D	52	dapa	2.5→5→1 0mg	400	58.00	55.30	89.60±21.40
			glipizide		401	59.00	54.90	90.50±22.60
Müller-Wieland 2018[69]	T2D	52	dapa	10mg	314	57.40	64.30	87.20±19.40
			dapa+saxagli ptin	10mg+5m g	312	59.20	60.90	88.00±19.60
			glimepiride		313	58.60	66.50	86.00±17.50
Cho 2018[70]	T2D	24	dapa	5mg	36	63.10	63.90	72.40±19.60
			pioglitazone		35	63.60	54.30	67.10±16.80
Scott 2018[71]	T2D	24	dapa	10mg	306	66.60	60.80	76.90±12.30
			sitagliptin		307	67.70	55.00	79.40±11.30
Shimizu 2018[72]	T2D	24	dapa	5mg	33	56.20	57.60	79.40±15.80
			control		24	57.10	62.50	76.90±19.00

Hayashi 2017[73]	T2D	12	dapa	5mg	40	53.80	80.00	86.20±18.40
			sitagliptin	50mg	40	54.20	75.00	83.50±22.70
Zinman 2015[74]	T2D	135	empa	10mg	2345	63.00	70.50	74.40±21.80
			empa	25mg	2342	63.20	71.90	74.30±21.10
			pbo		2333	63.20	72.00	74.00±21.10
Araki 2015[75]	T2D	52	empa	10mg	136	61.30	73.00	87.10±20.30
			empa	25mg	137	61.80	70.00	85.60±19.30
			metformin		63	60.00	75.00	86.90±14.80
Hollander 2019[76]	T2D	104	ertu	5mg	445	58.70	51.00	88.30±18.70
			ertu	15mg	435	58.00	43.90	86.60±18.30
			glimepiride		435	57.90	51.50	86.50±18.50
Tsurutani 2018[77]	T2D	12	ipra	50mg	60	53.50	68.30	77.20± /
			sitagliptin		59	54.00	67.80	80.40± /
baseline eGFR [45,60) ml/min/1.73m^2								
Takashima 2018[78]	T2D	52	cana	100mg	21	64.70	55.00	57.10±16.20
			pbo		21	65.40	60.00	55.40±12.30
Fioretto 2018[79]	T2D	24	dapa	10mg	160	65.30	56.90	53.48±8.76
			pbo		161	66.20	56.50	53.68±10.67
Pollock 2019[80]	T2D	24	dapa	10mg	145	64.70	70.00	50.20±13.00
			pbo		148	64.70	71.00	47.70±13.50
Barnett 2014[44]	T2D	52	empa	25mg	187	64.60	57.20	45.40±10.20
			pbo		187	65.10	56.70	44.30±10.30
Grunberger 2017[81]	T2D	26	ertu	5mg	158	66.70	53.20	46.80±7.80
			ertu	15mg	155	67.50	48.40	46.90±9.10

			pbo		154	67.50	46.80	46.00±9.40
Haneda 2016[82]	T2D	24	luseo	2.5mg	95	67.90	75.80	52.00±9.40
			pbo		50	68.40	78.00	52.40±8.20
baseline eGFR ＜45ml/min/1.73m^2								
Yale 2013[83]	T2D	26	cana	100mg	90	69.50	64.40	39.70±6.90
			cana	300mg	89	67.90	53.90	38.50±6.90
			pbo		90	68.20	63.30	40.10±6.80
Heerspink 2020[84]	CKD	52	dapa	10mg qd	2149	61.80	67.10	43.20±12.30
			pbo		2149	61.90	66.70	43.00±12.40
Bhatt 2021[85]	T2D	64	sota	200→400 mg	5292	69.00	55.70	44.40± /
			pbo		5292	69.00	54.50	44.70± /
Kohan 2013[86]	T2D	104	dapa	5mg qd	83	66.00	66.30	44.20±8.80
			dapa	10mg qd	85	68.00	65.90	43.90±10.60
			pbo		84	67.00	63.10	45.60±10.00
Barnett 2014[44]	T2D	52	empa	25mg	37	65.40	56.80	24.40±5.20
			pbo		37	62.90	51.40	22.00±4.40

Table S2. Evaluation for risk of bias in included RCTs

Author, year	Adequate randomization sequence generation	Adequate allocation concealment	Blinding of participants and caregivers	Blinding of outcome assessors and adjudicators	Free of infrequent missing outcome data	Free of selective outcome reporting	Free of other bias
Cusi,2018[1]	**Definitely yes** Using a computer-generated randomization schedule	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 7.7% (2/26) and 10% (3/30) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

Ji, 2015[2]	**Definitely yes** Using a computer-generated randomization schedule	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Definitely yes** There were 4.7% (21/450) and 4.0% (9/226) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Lavalle-González, 2013[3]	**Definitely yes** Using a computer-generated randomization schedule	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably no** There were 12.2% (90/735) and 15.3% (28/183) patients in SGLT2i and	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Wilding,2013[4]	**Definitely yes** Using a computer-generated randomization schedule	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Definitely no** There were 29.7% (93/313) and 42.3% (66/156) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Henry,2018[5]	**Definitely yes** Using an interactive voice response system/interactive web response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Definitely yes** There were 4% (2/50)and 2% (1/50)patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					across groups		
Jabbour, 2018[6]	**Definitely yes** Using an interactive voice response system and web-response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Definitely yes** There were 1.3% (3/228) and 1.3% (3/227) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Ji, 2014[7]	**Definitely yes** Using an interactive voice response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably no** There were 11.8% (31/261) and 14.4% (19/132) patients in SGLT2i and	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Mathieu, 2015[8]	**Definitely yes** Using an interactive voice response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 7.5% (12/160) and 4.4% (7/160) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					generally balanced across treatment groups, with similar reasons for missing data across groups		
Rosenstock,2015[9]	**Definitely yes** Using a centralized blocked randomization schedule	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 5.6% (10/179) and 8.5% (15/176) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					across groups		
Wilding, 2012[10]	**Definitely yes** Using a computer-generated randomization schedule	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably no** There were 14.5% (88/607) and 18.6% (36/193) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Fedinand,2019[11]	**Definitely yes** Using a computer-generated random sequence and a third-party	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 4.9% (4/82) and 8.9% (7/79) patients in SGLT2i and	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics

	interactive telephone/web-based system				control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		across groups
Hadjadj, 2016[12]	**Definitely yes** Using a computer-generated random sequence and an interactive voice and Web response system.	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 7.8% (53/680) and 11.7% (40/341) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					generally balanced across treatment groups, with similar reasons for missing data across groups		
Lewin, 2015[13]	**Definitely yes** Using a third-party interactive voice and web response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (details not reported)	**Definitely yes**	**Probably no** There were 15.8% (43/273) and 14.1% (19/135) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					across groups		
Ji,2019[14]	**Definitely yes** Using a computer-generated randomization code based on the method of random permuted blocks	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 7.1% (24/339) and 10.2% (17/167) patients in active agent and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Pratley,2018[15]	**Definitely yes** Using a an interactive voice response system/integrated	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 8.2% (40/488) and 10.5% (26/247) patients in active	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics

	web response system				agent and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		across groups
Rosenstock, 2018[16]	**Definitely yes** Using a computer-generated randomization code based on the method of random permuted blocks	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 5.1% (21/412) and 9.1% (19/209) patients in active agent and placebo groups with missing outcome data, respectively; missing outcome	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Inoue,2019[17]	**Definitely yes** Using a computer-generated random sequence	**Definitely yes** Open-labeled	**Definitely no** Open-labeled (participant, investigator)	**Definitely no**	**Probably yes** There were 8.3% (2/24) patients in SGLT-2 group with missing outcome data.	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Lu, 2016[18]	**Definitely yes** Using an interactive web randomization system with treatment codes	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (details not reported)	**Definitely yes**	**Probably no** There were 14.9% (13/87) and 16.9% (14/83) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Danne, 2018[19]	**Probably yes** Randomized, double-blind	**Probably yes** Randomized, double-blind	**Definitely yes** Double-blind (details not reported)	**Definitely yes**	**Probably no** There were 13.5% (71/524) and 12.8% (33/258) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					missing data across groups		
Grag, 2017[20]	**Probably yes** Randomized, double-blind	**Probably yes** Randomized, double-blind	**Definitely yes** Double-blind (details not reported)	**Definitely yes**	**Probably no** There were 13.6% (95/699) and 11.5% (81/703) patients in SGLT2i and control groups with missing outcome data respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Zambrowicz,2012[21]	**Definitely yes** Using a SAS-generated randomization	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant,	**Definitely yes**	**Probably yes** There was 8.3% (1/12) patient in placebo group	**Definitely yes**	**Probably yes** Generally balanced baseline

	schedule		investigator)		with missing outcome data.		characteristics across groups
Leiter 2014[22]	**Definitely yes** Using an interactive voice or web response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Definitely no** There were 31.2% (302/968) and 34.9% (168/482) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Handelsman 2018[23]	**Definitely yes** Using an interactive voice response	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind	**Definitely yes**	**Probably yes** There were 89.2% (411/468)	**Definitely yes**	**Probably yes** Generally balanced

	system		(participant, investigator)		patients in SGLT2i and control groups with missing outcome data, totally; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		baseline characteristics across groups
Jeon 2018[24]	**Probably yes**	**Probably yes** Open-labeled	**Definitely no** Open-labeled	**Definitely no**	**Probably yes**	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Ferrannini 2013[25]	**Probably yes**	**Probably yes** Open-labeled	**Definitely no** Open-labeled	**Definitely no**	**Probably yes** There were 8.03% (53/660) patients in SGLT2i and	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics

					control groups with missing outcome data, totally; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		across groups
Miller 2018[26]	**Definitely yes** Using a central electronic randomization system (interactive voice response system/integrated web response system)	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably no** There were 8.2% (16/194) and 21.6% (21/97) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					balanced across treatment groups, with similar reasons for missing data across groups		
Bode, 2015[27]	**Probably yes** Randomized, double-blind	**Probably yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Definitely no** There were 24.1% (79/477) and 33.3% (79/237) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

Forst, 2014[28]	**Definitely yes** Using an interactive voice response system/interactive web response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably no** There were 9.7% (22/227) and 20.9% (24/115) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Inagaki, 2013[29]	**Definitely yes** Using a block allocation method	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably no** There were 3.9% (12/307) and 13.3% (10/75) patients in SGLT2i and	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Inagaki, 2014[30]	**Definitely yes** Using a block allocation method	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably no** There were 6.7% (12/179) and 20.4% (19/93) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					generally balanced across treatment groups, with similar reasons for missing data across groups		
Inagaki, 2016[31]	**Definitely yes** Using a block allocation method	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Definitely yes** There were 0% (0/76) and 0% (0/70) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

Kadowaki, 2017[32]	**Probably yes** Randomized, double-blind	**Probably yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 4.3% (3/70) and 10.3% (7/68) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Rosenstock, 2016[33]	**Definitely yes** Using a computer-generated randomization schedule	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably no** There were 7.8% (37/474) and 13.5% (32/237) patients in SGLT2i and control groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Stenlöf, 2013[34]	**Probably yes** Randomized, double-blind	**Probably yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably no** There were 11.5% (45/392) and 16.7% (32/192) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were generally	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					balanced across treatment groups, with similar reasons for missing data across groups		
Araki, 2017[35]	**Definitely yes** Using an interactive web response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 9.0% (11/122) and 10% (6/60) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Bailey, 2015[36]	**Definitely yes**	**Definitely yes**	**Definitely**	**Definitely**	**Definitely no**	**Definitely**	**Probably yes**

	Using a computer-generated randomization by an interactive voice response system and were stratified by site in blocks of seven.	Randomized, double-blind	**yes** Double-blind (participant, investigator)	**yes**	There were 37.2% (74/199) and 44% (33/75) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**yes**	Generally balanced baseline characteristics across groups
Bolinder, 2012[37]	**Definitely yes** Using a computer-generated randomization schedule	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 8.8% (8/91) and 5.5% (5/91) patients in SGLT2i and placebo groups with missing	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		
McMurray, 2019[38]	**Definitely yes** Using an interactive voice or web response system	**Definitely yes** Randomized	**Probably yes** Randomized	**Probably yes**	**Probably yes** There were 10.5% (249/2373) and 10.9% (258/2371) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					generally balanced across treatment groups, with similar reasons for missing data across groups		
Schumm-Draeger, 2015[39]	**Definitely yes** Using an interactive web response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 7.4% (22/299) and 7.9% (8/101) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					across groups		
Strojek, 2011[40]	**Definitely yes** Using a computer-generated randomization schedule	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 8.2% (37/450)and 8.9% (13/146)patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Weber,2016[41]	**Definitely yes** Using a block allocation method	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 6.2% (14/225)and 9.8% (22/224)patients in SGLT2i and placebo groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Wilding, 2009[42]	**Probably yes** Randomized, double-blind	**Probably yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Definitely no** There were 8.3% (4/48) and 30.4% (7/23) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					treatment groups, with similar reasons for missing data across groups		
Yang, 2018[43]	**Definitely yes** Using an interactive voice response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 3.6% (5/139) and 6.8% (9/133) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Barnett, 2014[44]	**Definitely yes** Using a	**Definitely yes** Randomized,	**Definitely yes**	**Definitely yes**	**Probably yes** There were 9.2%	**Definitely yes**	**Probably yes** Generally

	computer-generated random sequence	double-blind	Double-blind (participant, investigator)		(18/195) and 10.3% (10/97) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		balanced baseline characteristics across groups
Haring, 2013[45]	**Definitely yes** Using a third-party interactive voice and web response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (details not reported)	**Definitely yes**	**Probably yes** There were 8.3% (37/444) and 10.7% (24/225) patients in SGLT2i and control groups with missing	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Haring, 2014[46]	**Definitely yes** Using a third-party interactive voice and web response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (details not reported)	**Definitely yes**	**Probably yes** There were 6.0% (26/431) and 10.1% (21/207) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					treatment groups, with similar reasons for missing data across groups		
Hattori,2018[47]	**Probably yes** Randomized	**Probably yes** Open-labeled	**Definitely no** Open-labeled (participant, investigator)	**Definitely no**	**Probably yes**	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Kario,2018[48]	**Definitely yes** Using a computer-generated randomization sequence	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Definitely yes** There were 1.6% (1/64) patients in placebo group with missing outcome data.	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Kadowaki, 2014[49]	**Definitely yes** Using a computer-generated random sequence and an interactive web response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 2.3% (10/438) and 8.3% (9/109) patients in SGLT2i and placebo groups with missing	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Kawamori,2018[50]	**Definitely yes** Using a computer-generated random sequence using a web-based interactive response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 2.7% (5/182) and 7.5% (7/93) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups,	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					with similar reasons for missing data across groups		
Kovacs, 2014[51]	**Definitely yes** Using a computer-generated random sequence and an interactive voice and web response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 6.9% (23/333)and 10.9% (18/165)patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Packer, 2020[52]	**Definitely yes** Using an	**Definitely yes** Randomized,	**Definitely yes**	**Definitely yes**	**Definitely yes** There were 1.2%	**Definitely yes**	**Probably yes** Generally

	interactive-response system that used a permuted-block design	double-blind	Double-blind (participant, investigator)		(22/1863) and 1.1% (20/1867) patients in active agent and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		balanced baseline characteristics across groups
Roden, 2013[53]	**Definitely yes** an interactive voice and internet-based response system, with a computer-generated random sequence, in block sizes of four	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably no** There were 8.5% (38/448)and 18.0% (41/228)patients in SGLT2i and placebo groups with missing	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Rosenstock, 2014[54]	**Definitely yes** Using a third-party interactive voice-and web-response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably no** There were 15.6% (59/377) and 16.9% (32/189) patients in SGLT-2i and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					treatment groups, with similar reasons for missing data across groups		
Rosenstock, 2015[55]	**Definitely yes** Using a third-party interactive voice and web response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Definitely no** There were 25.3% (82/324) and 30.6% (52/170) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Ross, 2015[56]	**Definitely yes**	**Definitely yes**	**Probably yes**	**Probably yes**	**Probably yes**	**Definitely**	**Probably yes**

	Using a block allocation method	Randomized,			There were 7.2% (63/876) and 3.7% (4/107) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**yes**	Generally balanced baseline characteristics across groups
Dagogo-Jack, 2018[57]	**Definitely yes** Using a computer-generated randomization schedule	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably no** There were 11.3% (35/311) and 14.4% (22/153) patients in active agent and placebo	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Terra,2017[58]	**Definitely yes** Using a computer-generated randomization code using the method of random permuted blocks	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Definitely no** There were 14.0% (43/308) and 22.2% (34/153) patients in active agent and placebo groups with missing outcome data, respectively; missing outcome	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Bando,2016[59]	**Definitely yes** Using a EDC system	**Definitely yes** Randomized, open-label	**Definitely no** open-label	**Definitely no**	**Definitely yes** All patients complete the study protocol without any withdrawals	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Han,2018[60]	**Definitely yes** Using a computer-generated randomization schedule	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Definitely no** There were 24.3% (18/74) and 17.4% (12/69) patients in SGLT-2 agent and placebo groups with missing outcome data, respectively;	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Ishihara, 2016[61]	**Probably yes** Randomized, double-blind	**Probably yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 3.4% (6/175) and 12.6% (11/87) patients in SGLT-2 agent and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups,	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					with similar reasons for missing data across groups		
Kashiwagi, 2015[62]	**Definitely yes** Using a block allocation method	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 10.1% (12/119) and 8.7% (4/46) patients in active agent and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Seino, 2018[63]	**Definitely yes** Using a	**Definitely yes** Randomized,	**Definitely yes**	**Definitely yes**	**Probably no** There were 1.9%	**Definitely yes**	**Probably yes** Generally

	computer-generated randomization schedule	double-blind	Double-blind (participant, investigator)		(3/159) and 13.5% (10/74) patients in SGLT-2 and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		balanced baseline characteristics across groups
Buse, 2018[64]	**Probably yes** Randomized, double-blind	**Probably yes** Randomized, double-blind	**Definitely yes** Double-blind (details not reported)	**Definitely yes**	**Probably no** There were 14.5% (76/525) and 18.7% (50/268) patients in SGLT2i and control groups with missing	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Kaku, 2014[65]	**Definitely yes** Using a central web-based system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably no** There were 7.3% (13/178) and 15.8% (9/57) patients in SGLT-2 and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					treatment groups, with similar reasons for missing data across groups		
Terauchi, 2017[66]	**Definitely yes** Using an interactive web response system (IWRS)	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Definitely yes** There were 2.8% (4/141) and 2.8% (2/70) patients in SGLT-2 and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Schernthaner 2013[67]	**Definitely yes** Using an interactive	**Definitely yes** Randomized,	**Definitely yes**	**Definitely yes**	**Definitely no** There were	**Definitely yes**	**Probably yes** Generally

	voice response system/interactive web response system	double-blind	Double-blind (participant, investigator)		32.8% (124/378) and 44.4% (168/378) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		balanced baseline characteristics across groups
Nauck 2011[68]	**Definitely yes** Using a predefined computer-generated randomization scheme	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Definitely no** There were 20.7% (84/406) and 23.0% (94/408) patients in SGLT2i and control groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Müller-Wieland 2018[69]	**Definitely yes** Using a computer-generated random sequence	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 7.5% (47/626) and 8.0% (25/313) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					balanced across treatment groups, with similar reasons for missing data across groups		
Cho 2018[70]	**Probably yes** Randomized	**Probably yes** Open-labeled	**Definitely no** Open-labeled (participant, investigator)	**Definitely no**	**Definitely yes** There were 0% (0/36) and 0% (0/35) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Scott 2018[71]	**Definitely yes**	**Definitely yes**	**Definitely**	**Definitely**	**Definitely yes**	**Definitely**	**Probably yes**

	Using an interactive voice response system	Randomized, double-blind	**yes** Double-blind (details not reported)	**yes**	There were 3.6% (11/307) and 2.6% (8/307) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**yes**	Generally balanced baseline characteristics across groups
Shimizu 2018[72]	**Probably yes** Randomized	**Probably yes** Open-labeled	**Definitely no** Open-labeled (participant, investigator)	**Definitely no**	**Probably no** There were 5.7% (2/35) and 14.3% (4/28) patients in SGLT2i and control groups with missing	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Hayashi 2017[73]	**Probably yes** Randomized	**Probably yes** Open-labeled	**Definitely no** Open-labeled (participant, investigator)	**Definitely no**	**Definitely yes** There were 0% (0/40) and 0% (0/40) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups,	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					with similar reasons for missing data across groups		
Zinman 2015[74]	**Definitely yes** Using a computer-generated random-sequence and interactive voice- and Web-response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (details not reported)	**Definitely yes**	**Definitely yes** There were 3.1% (144/4687) and 2.9% (67/2333) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Araki 2015[75]	**Definitely yes** Using a	**Definitely yes** Randomized	**Definitely no** Open-labeled	**Definitely no**	**Probably yes**	**Definitely yes**	**Probably yes** Generally

	computer-generated random sequence via an interactive web-based response system.		(participant, investigator)				balanced baseline characteristics across groups
Hollander 2019[76]	**Definitely yes** Using a computer-generated randomization schedule.	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Definitely no** There were 22.9% (202/881) and 24.8% (108/435) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

Tsurutani 2018[77]	**Probably yes** Randomized	**Probably yes** Open-labeled	**Definitely no** Open-labeled (details not reported)	**Definitely no**	**Definitely yes** There were 5% (3/60) and 3.4% (2/59) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Takashima,2018[78]	**Probably yes** Randomized	**Probably yes** Open-labeled	**Definitely no** Open-labeled (participant, investigator)	**Definitely no**	**Definitely yes** There were 4.8% (1/21) and 4.8% (1/21) patients in SGLT2i and control groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Fioretto,2018[79]	**Definitely yes** Using an interactive voice response system (IVRS) or interactive web response system (IWRS)	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 6.9% (11/160) and 9.3% (15/161) patients in SGLT-2 and placebo groups with missing outcome data, respectively; missing outcome data were generally	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					balanced across treatment groups, with similar reasons for missing data across groups		
Pollock, 2019[80]	**Definitely yes** Using the sponsor's interactive voice–web response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 9.3% (14/151) and 6.5% (10/153) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

Grunberger, 2017[81]	**Definitely yes** Using an interactive voice response system/integrated web response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably no** There were 10.5% (33/314) and 11.0% (17/154) patients in active agent and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Haneda, 2016[82]	**Probably yes** Randomized, double-blind	**Probably yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Probably yes** There were 6.3% (6/95) and 4% (2/50) patients in	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics

					SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups		across groups
Yale, 2013[83]	**Definitely yes** Using an interactive voice response system/interactive web response system.	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (details not reported)	**Definitely yes**	**Probably no** There were 12.3% (22/179) and 14.4% (13/90) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					data were generally balanced across treatment groups, with similar reasons for missing data across groups		
Heerspink, 2020[84]	**Definitely yes** Using an interactive voice-response or Web-response system	**Definitely yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Definitely yes** There were 0.47% (10/2152) and 0.23% (5/2152) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups

					missing data across groups		
Bhatt 2021[85]	**Probably yes** Randomized, double-blind	**Probably yes** Randomized, double-blind	**Definitely yes** Double-blind (participant, investigator)	**Definitely yes**	**Definitely yes** There were 1.1% (60/5292) and 1.5% (82/5292) patients in SGLT2i and control groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	**Definitely yes**	**Probably yes** Generally balanced baseline characteristics across groups
Kohan,2013[86]	**Probably yes** Randomized, double-blind	**Probably yes** Randomized, double-blind	**Definitely yes** Double-blind (details not	**Definitely yes**	**Definitely no** There were 42.9% (72/168) and 48.8%	**Definitely yes**	**Probably yes** Generally balanced baseline

		reported)	(41/84) patients in SGLT2i and placebo groups with missing outcome data, respectively; missing outcome data were generally balanced across treatment groups, with similar reasons for missing data across groups	characteristics across groups

Table S3. Subgroup analyses of efficacy among different drug categories.

Drug type	Renal function	Weighted mean difference	95%CI	P (over effect)	P (subgroup difference)
HbA1c (%)					
CANA	≥90	-0.77	-0.91, -0.64	＜0.001	
	60-89	-0.81	-0.93, -0.68	＜0.001	
	45-59	-0.20	-0.69, 0.29	0.42	
	＜45	NA	NA	NA	
	total	-0.79	-0.90, -0.68	＜0.001	0.06
DAPA	≥90	-0.49	-0.68, -0.30	＜0.001	
	60-89	-0.32	-0.45, -0.20	＜0.001	
	45-59	-0.27	-0.51, -0.04	0.02	
	＜45	-0.27	-0.97, 0.43	0.45	
	total	-0.36	-0.47, -0.25	＜0.001	0.42
EMPA	≥90	-0.32	-0.56, -0.07	0.01	
	60-89	-0.56	-0.65, -0.46	＜0.001	
	45-59	-0.44	-0.61, -0.27	＜0.001	
	＜45	0.48	-0.06, 1.02	0.08	
	total	-0.49	-0.57, -0.40	＜0.001	＜0.001
ERTU	≥90	-0.67	-0.85, -0.49	＜0.001	
	60-89	-0.60	-0.93, -0.26	＜0.001	
	45-59	-0.09	-0.24, 0.06	0.23	
	＜45	NA	NA	NA	
	total	-0.56	-0.76, -0.37	＜0.001	＜0.001
IPRA	≥90	-0.48	-0.69, -0.27	＜0.001	

	60-89	-0.80	-1.13, -0.48	＜0.001	
	45-59	NA	NA	NA	
	＜45	NA	NA	NA	
	total	-0.71	-0.99, -0.44	＜0.001	0.1
SOTA	≥90	-0.38	-0.52, -0.24	＜0.001	
	60-89	-0.28	-0.37, -0.19	＜0.001	
	45-59	NA	NA	NA	
	＜45	-0.43	-0.48, -0.38	＜0.001	
	total	-0.36	-0.43, -0.28	＜0.001	0.01
LUSEO	≥90	NA	NA	NA	
	60-89	-1.13	-1.34, -0.92	＜0.001	
	45-59	-0.2	-0.42, 0.02	0.08	
	＜45	NA	NA	NA	
	total	-0.67	-1.58, 0.25	0.15	＜0.001
TOFO	≥90	NA	NA	NA	
	60-89	-0.91	-1.03, -0.78	＜0.001	
	45-59	NA	NA	NA	
	＜45	NA	NA	NA	
	total	-0.91	-1.03, -0.78	＜0.001	NA
Weight (kg)					
CANA	≥90	-2.5	-3.05, -1.95	＜0.001	
	60-89	-1.73	-2.05, -1.42	＜0.001	
	45-59		NA	NA	
	＜45		NA	NA	
	total	-1.86	-2.22, -1.50	＜0.001	0.02

DAPA	≥90	-2.07	-3.43, -0.71	0.003	
	60-89	-1.97	-2.75, -1.19	＜0.001	
	45-59	-1.25	-2.08, -0.42	0.003	
	＜45	-3.35	-4.86, -1.85	＜0.001	
	total	-2.04	-2.74, -1.34	＜0.001	0.11
EMPA	≥90	-2.17	-2.64, -1.69	＜0.001	
	60-89	-1.93	-2.11, -1.75	＜0.001	
	45-59	-1.17	-1.75, -0.59	＜0.001	
	＜45	-1.00	-2.57, 0.57	0.21	
	total	-1.94	-2.11, -1.77	＜0.001	0.03
ERTU	≥90	-1.88	-2.12, -1.64	＜0.001	
	60-89	-2.56	-3.36, -1.75	＜0.001	
	45-59	NA	NA	NA	
	＜45	NA	NA	NA	
	total	-2.27	-2.74, -1.80	＜0.001	0.12
IPRA	≥90	-1.83	-3.12,	0.006	
	60-89	-1.66	-2.16, -1.15	＜0.001	
	45-59	NA	NA	NA	
	＜45	NA	NA	NA	
	total	-1.68	-2.11, -1.25	＜0.001	0.81
SOTA	≥90	-2.83	-3.14, -2.51	＜0.001	
	60-89	-3.73	-4.89, -2.57	＜0.001	
	45-59	NA	NA	NA	
	＜45	-1.16	-1.27, -1.05	＜0.001	
	total	-2.73	-3.82, -1.65	＜0.001	＜0.001

LUSEO	≥90	NA	NA	NA	
	60-89	-1.27	-1.69, -0.85	＜0.001	
	45-59	-1.27	-1.84, -0.7	＜0.001	
	＜45	NA	NA	NA	
	total	-1.27	-1.61, -0.93	＜0.001	1.00
TOFO	≥90	NA	NA	NA	
	60-89	-2.06	-2.69, -1.42	＜0.001	
	45-59	NA	NA	NA	
	＜45	NA	NA	NA	
	total	-2.06	-2.69, -1.42	＜0.001	NA
SBP (mmHg)					
CANA	≥90	-4.61	-5.77, -3.45	＜0.001	
	60-89	-4.58	-5.49, -3.67	＜0.001	
	45-59	-3.60	-4.90, -2.30	＜0.001	
	＜45	-5.95	-8.89, -3.01	＜0.001	
	total	-4.51	-5.18, -3.84	＜0.001	0.43
DAPA	≥90	-3.49	-5.24, -1.74	＜0.001	
	60-89	-2.75	-3.94, -1.56	＜0.001	
	45-59	-2.97	-6.09, 0.15	0.06	
	＜45	-5.61	-8.97, -2.24	0.001	
	total	-3.17	-4.37, -1.97	＜0.001	0.45
EMPA	≥90	-3.61	-4.72, -2.50	＜0.001	
	60-89	-3.84	-4.67, -3.02	＜0.001	
	45-59	-4.30	-7.13, -1.47	0.003	
	＜45	-12.20	-19.75, -4.65	0.002	

	total	-3.87	-4.55, -3.20	＜0.001	0.17
ERTU	≥90	-3.80	-4.77, -2.83	＜0.001	
	60-89	-3.84	-4.69, -2.98	＜0.001	
	45-59	NA	NA	NA	
	＜45	NA	NA	NA	
	total	-3.82	-4.46, -3.18	＜0.001	0.96
IPRA	≥90	-5.5	-9.56, -1.44	0.008	
	60-89	-2.23	-3.97, -0.50	0.01	
	45-59	NA	NA	NA	
	＜45	NA	NA	NA	
	total	-2.74	-4.33, -1.14	＜0.001	0.15
SOTA	≥90	-3.45	-4.36, -2.55	＜0.001	
	60-89	-3.59	-5.04, -2.15	＜0.001	
	45-59	NA	NA	NA	
	＜45	-2.44	-2.84, -2.04	＜0.001	
	total	-2.67	-3.02, -2.31	＜0.001	0.06
LUSEO	≥90	NA	NA	NA	
	60-89	-4.1	-7.48, -0.72	0.02	
	45-59	-2.6	-7.91, 2.71	0.34	
	＜45	NA	NA	NA	
	total	-3.67	-6.52, -0.82	0.01	0.64
TOFO	≥90	NA	NA	NA	
	60-89	-3.88	-6.04, -1.72	＜0.001	
	45-59	NA	NA	NA	
	＜45	NA	NA	NA	

	total	-3.88	-6.04, -1.72	＜0.001	NA
DBP (mmHg)					
CANA	≥90	-1.74	-2.37, -1.12	＜0.001	
	60-89	-1.83	-2.19, -1.47	＜0.001	
	45-59	-2.20	-3.01, -1.39	＜0.001	
	＜45	-1.65	-3.41, 0.11	0.07	
	total	-1.86	-2.14, -1.57	＜0.001	0.83
DAPA	≥90	-1.80	-2.02, -1.59	＜0.001	
	60-89	-0.73	-1.45, -0.01	0.05	
	45-59	NA	NA	NA	
	＜45	-0.47	-2.62, 1.68	0.67	
	total	-1.70	-1.91, -1.50	＜0.001	0.01
EMPA	≥90	-2.17	-2.88, -1.46	＜0.001	
	60-89	-1.70	-2.02, -1.38	＜0.001	
	45-59	-1.50	-3.20, 0.20	0.08	
	＜45	-5.70	-9.69, -1.71	0.005	
	total	-1.79	-2.08, -1.51	＜0.001	0.16
ERTU	≥90	-1.23	-1.83, -0.63	＜0.001	
	60-89	-1.36	-1.91, -0.81	＜0.001	
	45-59	NA	NA	NA	
	＜45	NA	NA	NA	
	total	-1.30	-1.70, -0.89	＜0.001	0.76
IPRA	≥90	-2.7	-5.32, -0.08	0.04	
	60-89	-1.10	-2.33, 0.14	0.08	
	45-59	NA	NA	NA	

	＜45	NA	NA	NA	
	total	-1.39	-2.51, -0.27	0.01	0.28
SOTA	≥90	-1.20	-1.81, -0.59	＜0.001	
	60-89	-1.90	-2.80, -1.00	＜0.001	
	45-59	NA	NA	NA	
	＜45	-0.80	-1.05, -0.55	＜0.001	
	total	-0.92	-1.15, -0.70	＜0.001	0.04
LUSEO	≥90	NA	NA	NA	
	60-89	-2.50	-4.84, -0.16	0.04	
	45-59	0.30	-3.30, 3.90	0.87	
	＜45	NA	NA	NA	
	total	-1.67	-3.63, 0.30	0.10	0.20
TOFO	≥90	NA	NA	NA	
	60-89	-2.76	-4.28, -1.24	＜0.001	
	45-59	NA	NA	NA	
	＜45	NA	NA	NA	
	total	-2.76	-4.28, -1.24	＜0.001	NA
eGFR (mL/min/1.73 m^2)					
CANA	≥90	-1.78	-5.90, 2.33	0.40	
	60-89	NA	NA	NA	
	45-59	4.10	0.75, 7.45	0.02	
	＜45	NA	NA	NA	
	total	0.12	-4.29, 4.54	0.96	0.03
DAPA	≥90	-1.99	-4.24, 0.27	0.08	
	60-89	0.05	-1.16, 1.25	0.94	

	45-59	NA	NA	NA	
	＜45	0.88	0.47, 1.29	＜0.001	
	total	-0.35	-1.24, 0.55	0.45	0.03
EMPA	≥90	1.15	-0.17, 2.46	0.09	
	60-89	-0.33	-0.90, 0.24	0.26	
	45-59	-2.50	-4.08, -0.92	0.002	
	＜45	-0.30	-2.99, 2.39	0.83	
	total	-0.26	-0.78, 0.26	0.33	0.007
ERTU	≥90	1.54	-0.43, 3.52	0.12	
	60-89	NA	NA	NA	
	45-59	NA	NA	NA	
	＜45	NA	NA	NA	
	total	1.54	-0.43, 3.52	0.12	NA
IPRA	≥90	8.42	-5.15, 21.99	0.22	
	60-89	-1.21	-2.44, 0.01	0.05	
	45-59	NA	NA	NA	
	＜45	NA	NA	NA	
	total	-1.14	-2.35, 0.08	0.07	0.17
SOTA	≥90	-0.78	-1.82, 0.25	0.14	
	60-89	-1.25	-2.66, 0.16	0.08	
	45-59	NA	NA	NA	
	＜45	-1.25	-1.51, -0.99	＜0.001	
	total	-1.22	-1.47, -0.97	＜0.001	0.69
LUSEO	≥90	NA	NA	NA	
	60-89	-1.30	-3.78, 1.18	0.30	

	45-59	-2.50	-4.30, -0.70	0.007	
	＜45	NA	NA	NA	
	total	-2.09	-3.54, -0.63	0.005	0.44
TOFO	≥90	NA	NA	NA	
	60-89	-2.87	-4.84, -0.89	0.004	
	45-59	NA	NA	NA	
	＜45	NA	NA	NA	
	total	-2.87	-4.84, -0.89	0.004	NA

HbA1c, glycosylated hemoglobin; SBP, systolic blood pressure; DBP, diastolic blood pressure; eGFR, estimated glomerular filtrationrate; CANA, canagliflozin; DAPA, dapagliflozin; EMPA, empagliflozin; ERTU, ertugliflozin; IPRA, ipragliflozin; LUSEO, luseogliflozin; SOTA, sotagliflozin; TOFO, tofogliflozin; NA, not available.

Table S4. Subgroup analyses of efficacy among follow-up period.

Follow-up period	Renal function	Weighted mean difference	95%CI	P (over effect)	P (subgroup difference)
HbA1c (%)					
＜1y	≥90	-0.59	-0.68, -0.50	＜0.001	
	60-89	-0.70	-0.79, -0.62	＜0.001	
	45-59	-0.16	-0.27, -0.05	0.005	
	＜45				
	total	-0.65	-0.71, -0.58	＜0.001	＜0.001
＞1y	≥90	-0.13	-0.28, 0.02	0.08	
	60-89	-0.26	-0.36, -0.16	＜0.001	
	45-59	-0.41	-0.57, -0.25	＜0.001	
	＜45	-0.13	-0.67, 0.42	0.65	
	total	-0.23	-0.32, -0.15	＜0.001	0.09
Weight (kg)					
＜1y	≥90	-2.09	-2.72, -1.47	＜0.001	
	60-89	-1.74	-1.89, -1.60	＜0.001	
	45-59	-1.26	-1.73, -0.8	＜0.001	
	＜45				
	total	-1.86	-2.17, -1.56	＜0.001	0.08
＞1y	≥90	-2.24	-2.86, -1.62	＜0.001	
	60-89	-2.67	-3.40, -1.94	＜0.001	
	45-59	-1.17	-1.75, -0.59	＜0.001	
	＜45	-1.88	-3.04, -0.72	0.001	
	total	-2.47	-3.01, -1.93	＜0.001	0.009

SBP (mmHg)					
＜1y	≥90	-3.95	-4.83, -3.06	＜0.001	
	60-89	-3.73	-4.23, -3.24	＜0.001	
	45-59	-2.87	-5.56, -0.19	0.04	
	＜45	-5.95	-8.89, -3.01	＜0.001	
	total	-3.87	-4.39, -3.36	＜0.001	0.44
＞1y	≥90	-3.68	-4.49, -2.87	＜0.001	
	60-89	-3.82	-4.75, -2.88	＜0.001	
	45-59	-3.72	-4.91, -2.54	＜0.001	
	＜45	-5.32	-8.96, -1.67	0.004	
	total	-3.79	-4.39, -3.19	＜0.001	0.86
DBP (mmHg)					
＜1y	≥90	-1.76	-1.95, -1.57	＜0.001	
	60-89	-1.52	-1.76, -1.27	＜0.001	
	45-59	0.30	-3.30, 3.90	0.87	
	＜45	-1.65	-3.41, 0.11	0.07	
	total	-1.66	-1.81, -1.52	＜0.001	0.33
＞1y	≥90	-1.46	-1.99, -0.94	＜0.001	
	60-89	-1.90	-2.31, -1.49	＜0.001	
	45-59	-2.07	-2.80, -1.34	＜0.001	
	＜45	-1.30	-2.95, 0.35	0.12	
	total	-1.71	-2.06, -1.37	＜0.001	0.44
eGFR (mL/min/1.73 m^2)					
＜1y	≥90	-0.19	-1.37, 0.98	0.75	
	60-89	-0.82	-1.27, -0.38	＜0.001	

	45-59	-2.50	-4.30, -0.70	0.007	
	＜45	NA	NA	NA	
	total	-0.77	-1.22, -0.32	＜0.001	0.11
＞1y	≥90	-1.02	-2.68, 0.63	0.23	
	60-89	0.24	-0.59, 1.08	0.57	
	45-59	0.63	-5.83, 7.09	0.85	
	＜45	-0.21	-1.77, 1.34	0.79	
	total	-0.06	-0.70, 0.58	0.85	0.60

HbA1c, glycosylated hemoglobin; SBP, systolic blood pressure; DBP, diastolic blood pressure; eGFR, estimated glomerular filtrationrate.

Table S5. Subgroup analyses of different AEs.

AEs	Renal function	OR	95%CI	P (over effect)	P (subgroup difference)
urinary tract infection	≥90	1.18	0.99, 1.39	0.06	
	60-89	1.06	0.97, 1.15	0.22	
	45-59	0.92	0.61, 1.38	0.68	
	＜45	1.05	0.94, 1.19	0.38	
	total	1.07	1.00, 1.14	0.04	0.60
genital tract infection	≥90	3.67	2.85, 4.73	＜0.001	
	60-89	3.95	3.31, 4.71	＜0.001	
	45-59	3.24	1.24, 8.46	0.02	
	＜45	2.83	2.04, 3.93	＜0.001	
	total	3.69	3.23, 4.20	＜0.001	0.37
amputation	≥90	1.48	0.06, 36.49	0.81	
	60-89	1.19	0.68, 2.09	0.54	
	45-59	3.08	0.12, 76.30	0.49	
	＜45	0.93	0.66, 1.30	0.67	
	total	1.01	0.76, 1.34	0.96	0.78
hypovolemia	≥90	1.65	1.08, 2.53	0.02	
	60-89	1.12	0.99, 1.26	0.06	
	45-59	1.65	0.82, 3.30	0.16	
	＜45	1.37	1.18, 1.60	＜0.001	
	total	1.24	1.13, 1.35	＜0.001	0.07
orthostatic hypotension	≥90	2.77	0.30, 25.75	0.37	
	60-89	1.23	0.54, 3.76	0.62	

	45-59	NA	NA	NA	
	＜45	1.52	0.06, 37.71	0.80	
	total	1.38	0.66, 2.89	0.39	0.80
bone fracture	≥90	0.59	0.29, 1.18	0.13	
	60-89	0.98	0.82, 1.17	0.81	
	45-59	0.36	0.11, 1.13	0.08	
	＜45	1.10	0.90, 1.35	0.35	
	total	0.99	0.87, 1.13	0.93	0.10
DKA	≥90	6.47	2.49,16.83	＜0.001	
	60-89	2.78	1.51, 5. 14	0.001	
	45-59	3.08	0.12, 76.30	0.49	
	＜45	1.27	0.80, 2.03	0.31	
	total	2.23	1.59, 3.11	＜0.001	0.01
hypoglycemia	≥90	1.07	0.57, 2.04	0.83	
	60-89	0.96	0.74, 1.23	0.72	
	45-59	0.94	0.69, 1.27	0.67	
	＜45	0.79	0.57, 1.11	0.17	
	total	0.96	0.78, 1.18	0.69	0.78

AEs, adverse effect; OR, odds ratio; DKA, diabetic ketoacidosis.

Table S6. Subgroup analyses of different AEs in placebo-controlled RCTs.

AEs	renal function	OR	95%CI	P (over effect)	P (subgroup difference)
urinary tract infection	⩾90	1.12	0.92, 1.36	0.25	
	60-89	1.03	0.94, 1.13	0.53	
	45-59	0.92	0.61, 1.38	0.68	
	＜45	1.05	0.94, 1.19	0.38	
	total	1.05	0.98, 1.12	0.20	0.80
genital tract infection	⩾90	3.27	2.47, 4.34	＜0.001	
	60-89	3.50	2.88, 4.25	＜0.001	
	45-59	3.24	1.24, 8.46	0.02	
	＜45	2.83	2.04, 3.93	＜0.001	
	total	3.32	2.88, 3.82	＜0.001	0.75
amputation	⩾90	1.48	0.06, 36.49	0.81	
	60-89	1.19	0.68, 2.09	0.54	
	45-59	3.08	0.12, 76.30	0.49	
	＜45	0.93	0.66, 1.30	0.67	
	total	1.01	0.76, 1.34	0.96	0.78
hypovolemia	⩾90	2.15	1.27, 3.64	0.005	
	60-89	1.10	0.98, 1.25	0.11	
	45-59	1.65	0.82, 3.30	0.16	
	＜45	1.37	1.18, 1.60	＜0.001	
	total	1.24	1.13, 1.35	＜0.001	0.02
orthostatic hypotension	⩾90	2.77	0.30, 25.75	0.37	
	60-89	1.28	0.52, 3.13	0.59	

	45-59	NA	NA	NA	
	＜45	1.52	0.06, 37.71	0.80	
	total	1.45	0.65, 3.23	0.36	0.82
bone fracture	⩾90	0.62	0.30, 1.26	0.19	
	60-89	0.97	0.81, 1.16	0.72	
	45-59	0.36	0.11, 1.14	0.08	
	＜45	1.13	0.75, 1.71	0.54	
	total	0.98	0.86, 1.12	0.78	0.18
DKA	⩾90	6.20	2.27, 16.91	＜0.001	
	60-89	2.44	1.26, 4.72	0.008	
	45-59	3.08	0.12, 76.30	0.49	
	＜45	1.00	0.33, 3.02	1.00	
	total	2.14	1.27, 3.61	0.004	0.12
hypoglycemia	⩾90	1.57	1.29, 1.91	＜0.001	
	60-89	1.07	0.98, 1.16	0.14	
	45-59	0.94	0.74, 1.19	0.60	
	＜45	0.80	0.62, 1.05	0.11	
	total	1.09	1.02, 1.17	0.02	＜0.001

AEs, adverse effect; RCTs, randomized controlled trials; OR, odds ratio; DKA, diabetic ketoacidosis.

Table S7. Efficacy of SGLT2i treatment in patients with different levels of renal function in placebo-controlled RCTs.

Efficacy outcomes	**Renal function**	**Mean difference**	**95%CI**	**P (over effect)**	**P (subgroup difference)**
HbA1c (%)	⩾90	-0.58	-0.67, -0.50	＜0.001	
	60-89	-0.67	-0.74, -0.60	＜0.001	
	45-59	-0.22	-0.35, -0.09	＜0.001	
	＜45	-0.13	-0.67, 0.42	0.65	
	total	-0.61	-0.67, -0.56	＜0.001	＜0.001
Weight (kg)	⩾90	-2.07	-2.31, -1.83	＜0.001	
	60-89	-1.81	-1.98, -1.64	＜0.001	
	45-59	-1.23	-1.59, -0.86	＜0.001	
	＜45	-1.88	-3.04, -0.72	0.001	
	total	-1.85	-1.99, -1.70	＜0.001	0.002
SBP (mmHg)	⩾90	-3.62	-4.09, -3.15	＜0.001	
	60-89	-3.67	-4.16, -3.17	＜0.001	
	45-59	-3.58	-4.67, -2.50	＜0.001	
	＜45	-5.36	-7.98, -2.74	＜0.001	
	total	-3.64	-3.99, -3.30	＜0.001	0.65
DBP (mmHg)	⩾90	-1.45	-1.75, -1.14	＜0.001	
	60-89	-1.58	-1.79, -1.36	＜0.001	
	45-59	-1.98	-2.69, -1.26	＜0.001	
	＜45	-0.83	-1.08, -0.58	＜0.001	
	total	-1.32	-1.46, -1.18	＜0.001	＜0.001
eGFR (mL/min/1.73	⩾90	-0.45	-1.39,0.50	0.35	

m^2)					
	60-89	-0.54	-1.00, -0.09	0.02	
	45-59	-0.68	-3.80, 2.45	0.67	
	＜45	-0.21	-1.77, -1.34	0.79	
	total	-0.54	-0.94, -0.14	0.008	0.98

HbA1c, glycosylated hemoglobin; SBP, systolic blood pressure; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate.

Table S8. Efficacy of SGLT2i treatment in patients with different levels of renal function (follow-up period ≥12 weeks)

Efficacy outcomes	Renal function	Mean difference	95%CI	P (over effect)	P(subgroup difference)
HbA1c (%)	≥90	-0.49	-0.59,-0.39	＜0.001	
	60-89	-0.58	-0.66,-0.5	＜0.001	
	45-59	-0.22	-0.35,-0.09	＜0.001	
	＜45	-0.13	-0.67,0.42	0.65	
	total	-0.53	-0.59,-0.47	＜0.001	＜0.001
Weight (kg)	≥90	-2.13	-2.66,-1.59	＜0.001	
	60-89	-2.06	-2.31,-1.82	＜0.001	
	45-59	-1.23	-1.59,-0.86	＜0.001	
	＜45	-1.88	-3.04,-0.72	0.001	
	total	-2.05	-2.32,-1.79	＜0.001	0.002
SBP (mmHg)	≥90	-3.83	-4.57,-3.09	＜0.001	
	60-89	-3.79	-4.24,-3.34	＜0.001	
	45-59	-3.58	-4.67,-2.5	＜0.001	
	＜45	-5.36	-7.98,-2.74	＜0.001	
	total	-3.86	-4.29,-3.42	＜0.001	0.68
DBP (mmHg)	≥90	-1.72	-1.9,-1.55	＜0.001	
	60-89	-1.63	-1.83,-1.43	＜0.001	
	45-59	-1.98	-2.69,-1.26	＜0.001	
	＜45	-0.83	-1.08,-0.58	＜0.001	
	total	-1.5	-1.62,-1.39	＜0.001	＜0.001
eGFR (mL/min/1.73 m^2)	≥90	-0.43	-1.4,0.54	0.38	
	60-89	-0.46	-0.93,0	0.05	

	45-59	-0.68	-3.8,2.45	0.67	
	＜45	-0.21	-1.77,1.34	0.79	
	total	-0.48	-0.87,-0.08	0.02	0.99

HbA1c, glycosylated hemoglobin; SBP, systolic blood pressure; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate.

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